

Allocating Influence: Government Advertising in Mexico's News Market

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Abstract

In many countries, governments are major media advertisers. Due to their prominent role, it is unclear whether they use their market power for political ends, and, if so, how they leverage that power. I study these questions in Mexico, where the government is one of the largest advertisers. Using novel data on advertising contracts and a comprehensive corpus of articles from national print media, I provide evidence of political favoritism. I use the 2018 transition from a right-wing to a left-wing federal administration and show that aligned outlets are 18.3 percentage points more likely to receive a contract. I benchmark these allocations against what a government would choose if it were buying audience reach efficiently or targeting its own voters. The observed allocation overpays outlets based on their ideology. The allocation of advertising to aligned media allows them to expand their newsroom capacity and their news production relative to misaligned outlets. The increase in production attracts more readers and raises circulation at aligned outlets. Lastly I show that the resulting circulation gains are driven mostly by the expansion in content rather than by shifts in reader preferences. Overall, these findings reveal a mechanism of media capture in competitive markets that requires neither censorship nor an exchange of coverage for revenue.

Keywords: Government advertising; Media capture; Partisan favoritism; Allocation efficiency; Media markets; Ideological alignment

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1 Introduction

Government advertising is a major source of revenue for media outlets in many developing and developed countries, particularly where private advertising markets are either limited or concentrated.¹ In such contexts, outlets may become financially dependent on public sector advertising. Much attention has focused on how this dependence might undermine editorial independence through quid pro quo arrangements, where outlets trade favorable coverage for advertising revenue (Di Tella and Franceschelli, 2011; Szeidl and Szucs, 2021).

However, in a competitive media market, such explicit arrangements are more costly and harder to sustain. When outlets compete for readers, any outlet that suppresses newsworthy information in exchange for government advertising risks losing its audience to rivals that cover it. This competitive discipline limits the scope for capture, because silencing information would require the government to negotiate with every outlet simultaneously, which is prohibitively expensive (Besley and Prat, 2006). I study a different mechanism through which the government can still shape the information environment in competitive markets. Rather than conditioning advertising on favorable coverage, the government uses its role as the dominant advertiser to selectively subsidize aligned outlets, allowing market competition to amplify the resulting asymmetry.

I focus on Mexico, where the federal government is one of the largest advertisers in a competitive newspaper market. While the stated purpose of government advertising is to promote public programs, highlight achievements, or disseminate prevention messages, its discretionary nature allows incumbents to reward supportive outlets or penalize critical ones. Such favoritism distorts media competition by directing resources toward ideologically aligned outlets. As a result, ideologically misaligned outlets are under-resourced, limiting their ability to expand their newsrooms or maintain coverage.

I first show that the government allocates advertising expenditures toward ideologically aligned outlets, both in the intensive and extensive margin. I implement a difference-in-differences specification that leverages the 2018 federal transition from a right-wing to a left-wing administration to compare changes in advertising spending between left- and right-wing outlets. I measure outlet ideology from text, using a penalized classifier trained on the editorials of five outlets with established ideological positions to place all outlets on a left-right dimension. Following the transition to a left-wing administration, left-wing outlets become 18.3 percentage points more likely to receive advertising contracts than right-wing outlets, a 33% increase relative to the pre-transition baseline probability of 0.56, which indicates substantial favoritism on the extensive margin.

Next, I show that governments reallocate advertising toward ideologically aligned outlets, and that this reflects favoritism toward the outlet's ideology rather than an effort to reach different demographic groups. One alternative explanation holds that what ministries advertise changed rather than whom they

¹Countries where the government is a major advertiser in the market include Bulgaria, Greece, Hungary, Poland, Romania, Slovakia, Slovenia, Albania, Macedonia, Serbia, Turkey, Argentina, Brazil, Colombia, Mexico, Peru, India, Pakistan, Myanmar.

favor. A left-wing government may promote more social programs whose intended beneficiaries overlap with the readership of left-wing outlets, which would make the reallocation look ideological even where no favoritism occurs. A second explanation is that the government prioritizes reaching its own voter base, if its supporters are disproportionately represented among readers of aligned outlets.

I address each explanation in turn. First, I restrict the analysis to advertising campaigns that ran under both administrations with the same content and the same stated demographic. Had ministries been pursuing particular demographic groups, their outlet choice should have remained stable. Yet even within these bipartisan campaigns, aligned outlets are 11.7 percentage points more likely to receive advertising and receive higher spending. A shift in what ministries advertise therefore cannot account for the reallocation.

Second, I construct two counterfactual benchmarks. Each solves an advertiser's problem and allocates a campaign's observed budget across newspapers to inform as many distinct readers as possible at each outlet's market advertising price. The first counts the readers in the campaign's stated demographic, which the legal cost-efficiency standard obliges ministries to buy. The second counts the government's predicted supporters, which an office-seeking politician would pursue. In the observed allocation aligned outlets receive \$21,609 more per campaign than the legal standard prescribes and \$23,932 more than voter targeting prescribes, worth 2.8 and 3.4 percent of a campaign budget, and they receive funding roughly 14 percentage points more often under either. Ministries also inform fewer readers than their budgets could have reached, delivering between 23 and 36 percent of what the benchmark would inform and falling short in all 383 campaigns. Under both benchmarks the excess spending goes to aligned outlets, and it rises with how closely an outlet aligns with the incumbent. Neither the campaign's stated demographic nor the incumbent's voter base produces that gradient.

By systematically channeling government advertising to aligned outlets while reducing funding to misaligned outlets, the government directly affects outlets' capacity to produce news. To examine these supply-side effects, I analyze key outcomes such as newsroom size and total article production. I implement the same difference-in-differences design that compares changes in these outcomes across aligned and misaligned outlets before and after the federal government transition, which allows me to identify differential effects on aligned versus misaligned outlets.

Reallocating government advertising creates asymmetric effects on newsroom capacity and content production. Following the transition to a left-wing government, aligned (left-wing) outlets experience substantial expansions, newsroom size increases by approximately 60% and news content production increases by approximately 42% compared to misaligned (right-wing) outlets. The increase in newsroom resources translates into greater production of original content across all hard news categories, namely politics, economics, and social policy. Quality measures including article length and wire-sourcing rates are statistically unchanged, indicating that the gains reflect a greater volume of original reporting rather than changes in quality per article. These asymmetric changes in production capacity reshape the information available to readers.

I then examine whether the expansion in content production translates into reader demand. Comparing circulation across left- and right-wing outlets before and after the 2018 transition, I find that left-wing outlets gain substantially in circulation relative to right-wing outlets. These gains could reflect two distinct forces, a supply-side effect, where more abundant content attracts new readers, or a demand-side effect, where the 2018 election shifted readers' political preferences toward left-wing media independently of any changes in content. To disentangle these channels, I exploit cross-state variation in the leftward shift in Morena's vote share between 2015 and 2018 and estimate a triple difference-in-differences specification. States where Morena's support grew more should exhibit a stronger preference-driven component, while content-driven gains should be present regardless of local political shifts. I find that the circulation gains are predominantly supply-driven, approximately 78% of the total gain is attributable to the expansion in newsroom capacity and content production, with the remaining 22% consistent with a preference-driven shift in readership.

This paper speaks to three literatures. The first is the theory of media capture. Besley and Prat (2006) show that pluralism raises the cost of capture, because a government seeking to suppress a story must buy the silence of every outlet capable of breaking it, and the price of that silence is each outlet's monopoly value. Later work relaxes the conditions under which that logic binds, Gehlbach and Sonin (2014) on state ownership and mobilization needs, Petrova (2008) on inequality and elite bribery, Petrova (2012) on the interaction between bias and audience size, Baron (2006) on career incentives inside the newsroom, and Prat (2018) on how influence should be measured when audiences are shared. This literature defines capture as influence over news content. I show that a government facing exactly the pluralism Besley and Prat (2006) identify as protective can still reshape the information environment without purchasing a single story, by determining newsroom capacity and market share. Competition does not weaken that instrument. Outlets convert a revenue subsidy into circulation only where they compete for an audience, so competition is what makes the instrument work.

The second is the empirical literature on government influence over the press. In autocracies the documented instruments are state ownership (Qin et al., 2018; Simonov and Rao, 2022) and censorship (Chen and Yang, 2019; Knight and Tribin, 2019). In democracies the documented mechanisms are bribes (McMillan and Zoido, 2002) and exchange. Di Tella and Franceschelli (2011) show that Argentine newspapers receiving more federal advertising devote less front-page space to corruption scandals; Szeidl and Szucs (2021) document two-way favors between the Hungarian government and connected outlets and separate owner ideology from favor exchange within a structural model; Durante and Knight (2012) show that partisan control of Italian public television shifted news content and that viewers responded by re-sorting across channels. DellaVigna et al. (2016) document the analogous private-sector channel, and Beattie et al. (2021) show that commercial advertisers can suppress unfavorable coverage. I document a channel through which a government uses advertising expenditures to shape the media environment in a competitive market, and that channel requires no favor exchange.

The third is the literature on how advertising revenue shapes news production. Angelucci and Cagé

(2019) show that the entry of television advertising in France cut newspaper revenue and led outlets to shed journalists and shift toward cheaper content; Cagé (2020) finds that competition among French local newspapers reduced information provision; Djourelova et al. (2025) trace how the displacement of classified advertising reshaped American local newspapers; and Petrova (2011) and Gentzkow et al. (2006) show that the historical growth of commercial advertising is what freed the American press from political subsidy in the first place. This work treats advertising revenue as a market outcome that happens to have political consequences. I treat it as a policy instrument. The same revenue channel that these papers show governs newsroom investment is, in my setting, allocated at the discretion of the incumbent and conditioned on the recipient's ideology, which converts a market force into a mechanism of political influence and reverses the historical relationship that Petrova (2011) describes.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting for federal advertising in Mexico. Section 3 presents the data and the measurement of outlet ideology. Section 4 documents ideological favoritism in allocation. Section 5 constructs the cost-efficiency and voter-targeting benchmarks. Section 6 traces the downstream effects on newsroom capacity, content production, and circulation. Section 7 concludes.

2 Institutional Background

Mexico's press-government relationship has shifted across instruments but not in its underlying logic. From 1935 to 1990, the government controlled newsprint supply through PIPSA, a state-owned company responsible for the bulk of newsprint supply to the press through production and importation, punishing critical outlets by withholding or degrading paper deliveries. Direct cash payments to journalists ran alongside PIPSA as a parallel instrument throughout the PRI era. When PIPSA was privatized in 1998 and the democratic transition of 2000 made these instruments politically costly, formal advertising contracts took their place. The shift promised outlets a steady and predictable revenue stream to replace those arbitrary cash payments. In practice, it created a new form of dependence. Outlets came to rely on government advertising for a large share of their total revenue, and the government gained the power to award or withhold contracts, a lever far more institutionalized than what it had replaced.

Advertising contracts are allocated to outlets by federal ministries through a system of annual campaign programs and formal guidelines. Federal advertising campaigns disseminate information about government policies and achievements, social programs, prevention and awareness initiatives, and civic and cultural obligations. For citizens, they are the primary channel through which they learn about programs they are eligible for, public health risks, and their rights and obligations. For the government, they serve as a vehicle for communicating the administration's policy record and building public awareness of its work. For each advertising campaign, a ministry specifies a communication objective, such as promoting a social program or informing citizens about a public service, along with a target demographic defined by age,

income, and education level. Each ministry presents its annual advertising budget to Congress for approval at the start of each fiscal year. Outlets seeking federal contracts must register with the National Registry of Printed Media, which records each outlet's circulation, readership demographics, and advertising prices. The registry provides all the inputs a ministry would need to determine efficient allocations. No independent body audits allocation decisions, and deviating from what the registry data would justify carries no formal penalty. These guidelines apply to communication campaigns, which account for 95 percent of total federal advertising expenditure. General announcements, which cover routine operational notices unrelated to government policy, carry no requirement to specify audiences or objectives and are subject to no oversight.

Mexico's print media market expanded and fragmented alongside the democratic opening of the 1990s. As single-party dominance eroded, new newspapers entered national circulation, and by the 2000 transition more than a dozen daily titles competed for the same readers and advertisers. The private advertising market grew alongside circulation but remained thin. Government contracts represented a large share of advertising revenue at most outlets, and commercial sources could not replace what exclusion from the official registry would cost. The sample period captures this fragmented structure. The Herfindahl-Hirschman Index averages 679 across 2014–2022, and in each year at least 39 percent of outlets are needed to account for half of total circulation. In this market, the government shapes the information environment not by suppressing content but by determining which outlets have the resources to produce it.

3 Data and Measurement

3.1 Data Sources

I construct a novel panel linking three datasets, government advertising contracts from federal administrative records, outlet-level variables from the National Print Media Registry, and newspaper content collected from printed editions.

Government advertising contracts. I draw administrative data from the Ministry of the Interior's Social Communication Strategy program, which records all federal advertising contracts between 2014 and 2022. Each contract identifies the ministry, the recipient outlet, the date, the amount spent, the number of advertisements by size, and the price paid per advertisement. For each campaign, ministries also report the communication goal, the target demographic (income, education, age, and gender), and whether it promotes government policies and achievements, social programs, prevention and awareness, or civic and cultural obligations.²

Outlet-level variables. Every outlet seeking federal advertising must register with the National Print Media Registry and update its information each year. The registry covers approximately 300 outlets,

²I obtain campaign characteristics through information requests to federal ministries via the National Transparency Platform. By law, government entities must provide public information requested by citizens through this platform.

including both national and local outlets. For each outlet, the registry records the average daily circulation at the state level, the share of readers by socioeconomic level (high, middle, and low), gender (female and male), education (basic, secondary, and advanced), age group, and the legal name of the outlet's owner. Additionally, outlets report list prices per cm² for advertising space across sections and page locations; I average these to construct a single market price per outlet. I restrict the sample to national outlets that publish news daily (or at least five days per week), dropping magazines and outlets focused on entertainment or sports. This leaves approximately 25 national outlets per year. I further limit the analysis to 18 outlets for which complete data on daily articles and historical registry information are available. On average, these 18 outlets account for approximately 82% of national print circulation each year.

News content. I collect the text of all print issues for the 18 national outlets from 2014 to 2022, totaling approximately 6 million articles.³ From each issue, I extract article text, observed advertisements, bylines, and publication dates. I classify advertisements as public (federal, state, legislative, or judicial) or private. I measure newsroom size from bylines, distinguishing wire sources (Reuters, Associated Press, Notimex), house staff, and individual journalists.

Table 1 reports summary statistics for the full sample, separately for the right-wing (PRI) and left-wing (Morena) administrations. Government advertising falls from \$1.7 million per outlet under PRI to \$467,000 under Morena, a decline that, if distributed unevenly across outlet types, translates into revenue shocks that differ by outlet ideology. Even at these levels, government advertising accounts for roughly one-third of all advertising space in the average outlet, with a standard deviation of 17 percentage points indicating that some outlets depend far more heavily on public contracts than others. The same discretion that governs outlet selection shapes the prices ministries pay, the government pays approximately twice the market rate, \$21.93 per cm² under PRI against a market price of \$9.00, and \$13.41 under Morena against \$8.28. The average outlet holds a 5.6% market share, consistent with a market where no outlet dominates. Together, these patterns, a government advertiser that accounts for one-third of ad space, prices that exceed market rates, and allocation decisions that no independent body audits, describe a setting where government choices reshape the market.

3.2 Measuring Ideological Alignment

Identifying ideological favoritism requires a measure of where each outlet stands on the left-right dimension. I construct one from text, following the logic of Gentzkow and Shapiro (2010) that the differential use of partisan language identifies ideological position, and using the penalized estimation of Gentzkow et al. (2019) to recover a sparse and interpretable set of partisan phrases. I estimate the measure on editorial board pieces rather than news content, because editorials reflect the outlet's institutional stance, whereas news coverage shifts with the topics an outlet covers in a given period.

Fifteen of the eighteen outlets publish a dedicated editorial section, and I use the 8,938 editorials they

³See Table A4 for per-outlet totals and yearly averages.

Table 1: **Summary Statistics**

	Right-Wing Government (PRI) (2012–2018)		Left-Wing Government (Morena) (2018–2024)	
	Mean	SD	Mean	SD
<i>Panel A: Government Advertising</i>				
Gov. advertising per outlet (USD, annual)	1,709,158	1,769,201	466,711	826,105
Price paid by government (USD per cm ²)	21.93	16.90	13.41	12.10
Share of public advertising (%)	34.1	17.0	33.8	16.9
Number of ministries contracting	11.0	5.9	6.0	3.8
<i>Panel B: Outlet Characteristics</i>				
Daily circulation (thousands)	115.1	56.6	115.7	57.8
Average market share (%)	5.7	2.8	5.6	2.7
Monthly articles	2,497	1,625	2,275	1,273
Newsroom size	198	160	204	139
Market advertising price (USD per cm ²)	9.00	8.35	8.28	7.41

Outlet-year observations. The sample covers 18 national newspapers from 2014 to 2022. USD values in 2022 constant dollars. Gov. advertising per outlet aggregates all federal ministry contracts per outlet and year. Share of public advertising is the fraction of observed advertisements classified as government-sourced from newspaper content. Market advertising price is the average list price per cm².

published between 2014 and 2017. Restricting the estimation window to the pre-transition period keeps the measure from absorbing vocabulary specific to the incoming administration, so that the recovered dimension is left-right rather than support for or opposition to the 2018 winner, and it makes outlet ideology a pre-treatment characteristic. Each editorial is lowercased, stripped of accents, digits, and punctuation, cleared of Spanish stopwords, lemmatized, and then stemmed, with tokens shorter than three characters dropped. Phrases are the unigrams and bigrams formed from the resulting stems.

Two outlets serve as left-wing anchors, La Jornada and La Prensa, and three as right-wing anchors, Reforma, El Universal, and El Economista. I select these five based on their editorial histories and established positions in the Mexican media landscape. The estimation vocabulary is built from the 3,836 editorials these five outlets published in the window, subject to three restrictions, a phrase must appear in at least three of the five anchor outlets, so that no single outlet’s recurring vocabulary can define the dimension; it must appear in at least five anchor editorials; and it must not contain an outlet name, a column title, or a formatting artifact.

I represent each anchor editorial as a vector of phrase proportions that sums to one, and I fit an L_1 -penalized logistic regression of an indicator for right-wing anchor authorship on that vector. The penalty is selected by scanning sixty values of the regularization parameter and retaining the solution closest to a target of five hundred non-zero phrases, following the sparsity target in Gentzkow et al. (2019). The selected model retains 514 phrases, 224 of which load right-wing and 290 left-wing. Table 2 reports a subset chosen for interpretability. Phrases loading left-wing include pension, Cuba, promise, countryman, teacher, while phrases loading right-wing include investment, institution, economic growth, competence, professor.

Each article receives a score equal to the inner product of the estimated loadings with its own vector of phrase proportions, and each outlet’s score is the mean of that quantity across its articles. I subtract the midpoint between the left- and right-wing anchor group means, so that zero falls equidistant between

the two clusters, and I classify an outlet as left-wing when its recentered score is negative and right-wing otherwise. The five anchors are scored by the same rule as every other outlet rather than being held at their training labels.

Diario de México, El Herald de México, and Esto publish no editorial section, so I score them by applying the estimated loadings to their news articles under the same preprocessing, scoring rule, and midpoint. Appendix B.1 resamples each of these three corpora down to the size of a typical editorial corpus and reports the resulting distribution of scores, which is stable in every draw.

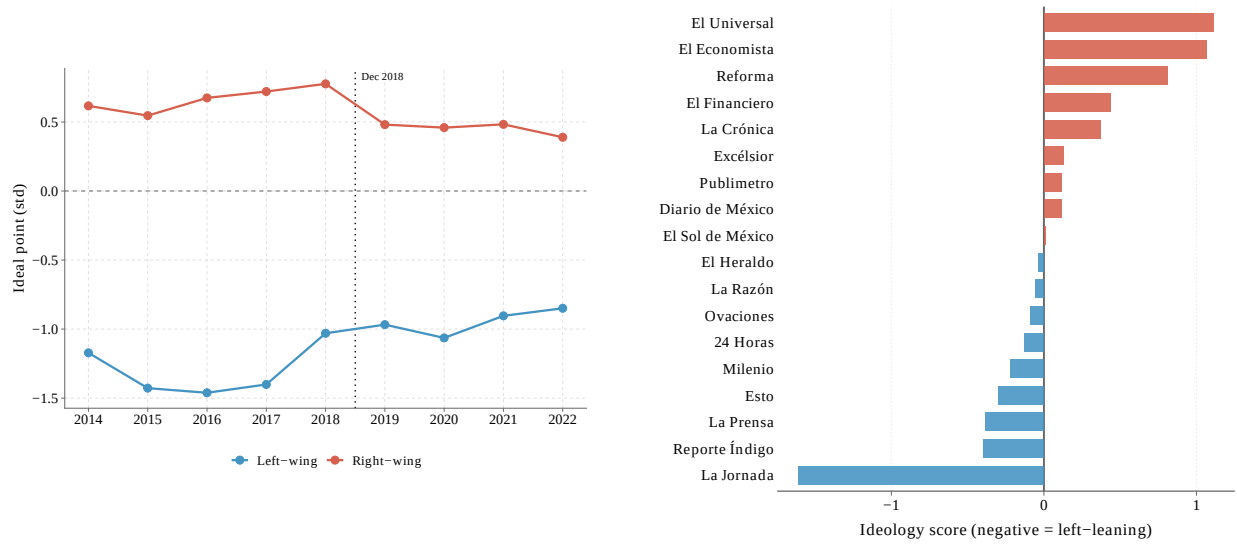
Table 2: **Top Partisan Phrases**

<i>Right-wing Phrases</i>			
American (estadounidens)	Capacity (capac)	Official (funcionari)	Local (local)
Competence (competent)	Union (sindicat)	Investment (inversion)	Structure (estructur)
Professor (profesor)	Institution (institucion)	Org. crime (crim organiz)	Internet (internet)
Transformation (transform)	Corruption (corrupcion)	Citizenship (ciudadani)	Process (proces)
State-run (estatal)	Service (servici)	Opportunity (oportun)	Infrastructure (infraestructur)
Freedom (libert)	Risk (riesg)	Institutional (institucional)	Poll (encuest)
Criminal (criminal)	Growth (crecimient)	Fiscal (fiscal)	
<i>Left-wing Phrases</i>			
Tax auth. (sat)	Injustice (injust)	Environmental (ambiental)	Hunger (hambr)
War (guerr)	Ministry (secretari)	Gas hike (gasolinaz)	Common sense (sent comun)
People (puebl)	Att. General (procurador)	Pollution (contamin)	Teacher (maestr)
Electoral roll (padron)	Promise (promes)	Countryman (paisan)	Disarmament (desarm)
Government (govern)	Gov. official (mandatari)	Odebrecht (odebrecht)	Misery (miseri)
Ayotzinapa (ayotzinap)	Femicide (femicididi)	Health (salud)	Electoral process (proces electoral)
Pension (pension)	Cuba (cub)	North American (norteamerican)	

The L_1 -penalized logistic regression retains 514 phrases with non-zero loadings, 224 loading right-wing and 290 loading left-wing. This table reports 55 of them, selected for interpretability rather than by the rank of the loading. Phrases are shown in English with the Spanish stem in parentheses.

The right panel of Figure 1 plots each outlet’s recentered score, and this classification remains fixed throughout the analysis. Nine outlets fall on each side. Several score close to the midpoint, among them the four with the thinnest editorial corpora. Ovaciones, Publimetro, El Sol de México, and La Razón each contribute fewer than forty editorials, which motivates the robustness checks using a continuous score and a three-category measure reported in Appendix B. The left panel applies the same phrase loadings to editorials published in each year from 2014 to 2022, for the fifteen outlets with editorial sections. The two groups remain clearly separated across the full period and their ordering never reverses, indicating that the classification estimated on the pre-transition window is not specific to that window.

Figure 1: Outlet Ideology Scores and Stability (2014–2022)

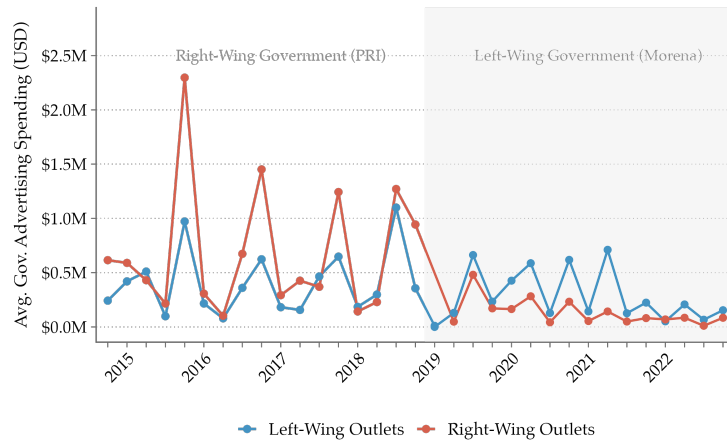


Phrase loadings are estimated by L_1 -penalized logistic regression on the editorials of the five anchor outlets. The right panel plots the score of each of the 18 outlets, recentered so that zero lies equidistant between the left- and right-wing anchor group means. Negative values indicate left-leaning positions and positive values indicate right-leaning positions. The left panel applies the same loadings to editorials published in each year and plots the article-weighted group mean.

3.3 Motivating Patterns

I document descriptive patterns in government advertising allocation, before turning to the causal analysis. Figure 2 plots the time series of government advertising expenditures across left- and right-wing newspapers among the 18 national circulating newspapers from the sample.

Figure 2: Government Advertising Spending by Outlet Ideology (2014–2022)



This figure displays the average quarterly advertising spending per outlet by ideological alignment. The sample includes 9 left-wing outlets and 9 right-wing outlets. Spending values are adjusted to 2022 constant pesos and converted to USD using the 2022 average exchange rate. The sample includes advertising from communication campaigns from government ministries. Average spending is calculated by summing total spending per outlet-quarter within each ideological group, then averaging across outlets. The vertical dashed line marks the presidential transition in December 2018 (Q1 2019), separating the PRI administration (2014–2018) from the Morena administration (2019–2022).

Under the right-wing PRI administration, right-wing outlets received more government advertising per quarter than left-wing outlets. Following the December 2018 transition to Morena, this pattern reverses.

Spending shifts toward left-wing outlets while right-wing outlets see a decline. Immediately after the transition, total spending drops sharply across all outlets, consistent with the austerity the incoming government announced, before stabilizing at the lower levels shown in Table 1.

The raw patterns are consistent with ideological favoritism but do not establish it causally. The post-2018 increase in spending to left-wing outlets could reflect changes in their audience size, market position, or the mix of campaigns the new government chose to run. Section 4 isolates the role of ideological alignment by exploiting variation across the transition within outlet–ministry pairs.

4 Ideological Favoritism in Advertising Allocation

Governments may prefer to advertise in outlets that already provide favorable coverage, either because they want to reach supportive audiences or because of shared ideology. To test whether such affinity shapes allocation decisions, I examine whether changes in government lead to shifts in spending toward ideologically aligned outlets.

I employ a difference-in-differences specification that leverages variation in government transitions and the ideological alignment of newspapers. I focus on the presidential transition in 2018, where the left-wing party Morena came to power, replacing the right-wing party PRI. Newspaper ideological positions are fixed over time and inferred from the text-based measures described in Section 3.2. I estimate the following regression:

$$Y_{ngt} = \beta \text{Left-Wing Gov}_t \times \text{Left-Wing Outlet}_n + \mu_t + \delta_{ng} + \epsilon_{ngt}, \quad (1)$$

where Y_{ngt} represents one of three outcome variables of government advertising to outlet n from ministry g in year t . First, an indicator for whether the outlet receives any advertising contract (extensive margin). Second, the level of total advertising spending in 2022 USD, including zero values for outlet–ministry pairs with no contracts in a given year (intensive margin). Third, conditional on receiving advertising, the price markup paid by the government relative to the outlet’s average market price, computed as (government price – market price) / market price, which captures whether the government pays above-market rates to advertise in ideologically aligned outlets.⁴ The variable Left-Wing Gov_t is an indicator equal to 1 when the presidency is held by a left-wing party, and $\text{Left-Wing Outlet}_n$ equals 1 if the outlet is classified as left-leaning and 0 if right-leaning.

Without ministry fixed effects, the estimates could reflect cross-ministry variation in budget trajectories, a ministry that structurally favors left-wing outlets for non-ideological reasons (e.g., because its target audience skews left) may also have expanded its budget under the left-wing government, generating a

⁴Market prices are constructed from outlets’ published list prices in the National Print Media Registry, which report the price per cm² by advertisement size, page location, section, and color. For each outlet-year, I predict the market price per page. For government contracts, I observe the price paid per advertisement and its size, from which I calculate the government price per page. Both prices are normalized by total circulation to express them per reader, and the markup is computed as (government price – market price) / market price. A positive markup indicates the government paid above-market rates, which may reflect purchases of more prominent placements (e.g., front page, color, premium sections).

spurious correlation. δ_{ng} removes this confound by identifying β from within-pair changes. Hence, the panel is structured at the outlet $\times$ ministry $\times$ year level. The specification also includes year fixed effects (μ_t) to absorb shocks common to all outlets in a given year, including the aggregate path of the federal communication budget. Standard errors are clustered at the newspaper level, as the treatment variable is defined at the newspaper level and errors across all ministry relationships of the same newspaper are correlated by construction. The identification strategy relies on the common trends assumption, in the absence of a shift in the party in power, advertising allocations to left- and right-leaning outlets would have followed common trends.

Table 3 presents the results. Columns (1)–(3) use the full sample of all advertising campaigns, restricted to the 10 ministries that ran campaigns across both administrations.⁵ These columns report the extensive margin, the intensive margin in levels, and the price markup.

Higher spending on aligned outlets could reflect ideology, or it could be rationalized by audience composition, if the government allocates more to left-wing outlets because those outlets reach its supporting audiences. To disentangle these explanations, I restrict Columns (4)–(6) to bipartisan campaigns, programs that ran under both administrations with identical stated target demographics, so that any differential allocation toward aligned outlets cannot be attributed to differences in whom the government was trying to reach. Columns (1)–(2) and (4)–(5), estimated at the outlet–ministry–year level, have identical numbers of observations, making the two panels directly comparable. Columns (3) and (6) report the price markup, estimated at the campaign–outlet level using the full cross-ministry panel, and therefore have a different number of observations.

Table 3: Differential Effects of Government Partisanship on Advertising Spending Across Left- and Right-Wing Outlets

	All Campaigns			Bipartisan Campaigns		
	Prob(Ads > 0)	Level (2022 USD)	Price Markup	Prob(Ads > 0)	Level (2022 USD)	Price Markup
	(1)	(2)	(3)	(4)	(5)	(6)
Left-Wing Government $\times$ Left-Wing Outlet	0.183** (0.082)	105,362* (53,615)	1.481 (1.314)	0.117* (0.056)	15,913* (9,017)	0.557 (0.896)
Mean (pre-2018, left)	0.56	107,562	4.67	0.26	7,563	2.06
Mean (pre-2018, right)	0.78	160,559	7.20	0.36	16,484	2.59
Year FE	✓	✓	✓	✓	✓	✓
Outlet $\times$ Ministry FE	✓	✓		✓	✓	
Agency $\times$ Outlet FE			✓			✓
Campaign FE			✓			✓
Observations	1,410	1,410	2,617	1,410	1,410	485
R ²	0.58	0.58	0.36	0.41	0.16	0.68

Note: *p<0.1; **p<0.05; ***p<0.01. SEs clustered at the outlet level. Cols. (1) and (4): indicator for positive advertising spending (extensive margin). Cols. (2) and (5): level of advertising spending in 2022 USD (intensive margin, including zeros); both restrict to the 10 ministries with bipartisan campaigns so that observations are comparable across panels ($N = 1,410$). Cols. (3) and (6): price markup (government price relative to average market price), estimated at the campaign–outlet level with agency–outlet and campaign fixed effects; these columns use the full campaign–outlet panel across all ministries. The Bipartisan Campaigns panel restricts to the 17 campaigns implemented across both administrations with identical target demographics. Means computed over the pre-2018 period.

Column (1) shows that left-wing outlets are 18.3 percentage points more likely to receive advertising from

⁵The 10 ministries are Culture, Defense, Education, Energy, Foreign Affairs, Health, Interior, Navy, Presidency, and Tourism.

a given ministry following the transition, a 33% increase relative to the pre-transition baseline probability of 0.56. Beyond selection, Column (3) reports the price markup paid by the government relative to the outlet's average market rate. The market price benchmark is the outlet's average list price across all placement types, so the estimate conflates two distinct channels, price discrimination, whereby the government pays above the market rate for equivalent placements in aligned outlets, and a composition effect, whereby the government systematically selects more prominent and costlier placements, such as front-page or color advertisements, in those outlets.

Because treatment affects the extensive margin, Column (1) documents a substantial shift in whether ministry–outlet pairs receive any advertising. Using a log-like transformation such as the inverse hyperbolic sine to measure the intensive margin would yield a coefficient that is not interpretable as a percentage change and sensitive to any linear rescaling of the outcome, such that the coefficient changes depending on whether spending is expressed in dollars, thousands of dollars, or pesos (Chen and Roth, 2024). I report the intensive margin in levels (Column 2), which is the unconditional average treatment effect in 2022 USD. This parameter is well-defined and directly interpretable; it averages over ministry–outlet pairs that begin receiving advertising after the transition (extensive margin) and pairs that received positive advertising under both administrations (intensive margin).

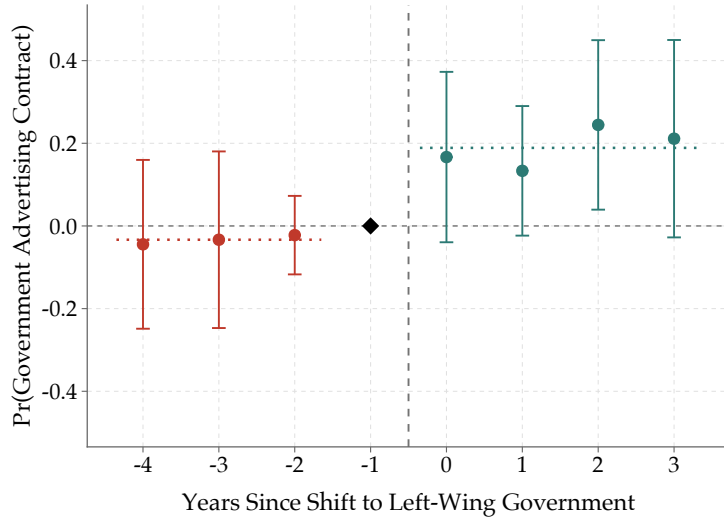
Aligned outlets receive \$105,362 more per ministry per year than misaligned outlets, approximately 0.36 standard deviations of the outcome. Both groups lost advertising revenue in absolute terms, because the incoming administration cut total federal advertising across these ministries by 48 percent. Spending on aligned outlets fell by 9 percent across the transition, while spending on misaligned outlets fell by 72 percent. The estimate therefore measures the extent to which aligned outlets were shielded from a contraction that fell almost entirely on their competitors, rather than an increase in what they received. Appendix C provides three robustness exercises following Chen and Roth (2024): (i) the inverse hyperbolic sine specification for comparability with prior literature; (ii) a Poisson regression, which is scale-invariant and whose coefficient is interpretable as a semi-elasticity; and (iii) Lee bounds (Lee, 2009) that separately identify the conditional intensive margin effect for the subpopulation of ministry–outlet pairs that receive positive advertising under both administrations.

Columns (4)–(6) disentangle ideological favoritism from audience-driven allocations by restricting the sample to bipartisan campaigns. Aligned outlets are still 11.7 percentage points more likely to receive advertising, a 45% increase relative to their pre-transition baseline of 0.26, and receive \$15,913 more per ministry per year, confirming that favoritism drives the reallocation independently of audience composition.

To assess the validity of the parallel trends assumption, Figure 3 presents event study estimates corresponding to Column (1) of Table 3. Event time is defined as 12-month periods relative to the December 2018 transition, with the period immediately preceding the transition (December 2017–November 2018) normalized to zero. Following the transition, the coefficients become positive and statistically significant, indicating a sharp increase in advertising spending to left-wing outlets relative to right-wing outlets, and

the effect grows persistently through 2022.

Figure 3: Event Study: Extensive Margin of Government Advertising by Outlet Ideology



Coefficients on the interaction between event time and a left-wing outlet indicator, from Column (1) of Table 3. Reference group: right-wing outlets. Omitted period: December 2017–November 2018. Error bars show 95% confidence intervals; standard errors clustered at the outlet level. Outlet–ministry fixed effects included.

The flat pre-transition coefficients and the increasing post-transition estimates are consistent with a discontinuous change in advertising allocations driven by the party transition, rather than pre-existing differences in trajectory. Event study plots for other specifications are presented in Appendix Figure C1.

5 Counterfactual Allocations

Section 4 shows that ministries allocate advertising to ideologically aligned outlets. A government that targets an audience would produce the same pattern, whether it targets the campaign’s stated demographic or its own voter base. I construct two counterfactual benchmarks, each allocating a campaign’s observed budget as a government would if it cared only about reaching one of those audiences, and I compare both with what ministries spent. The benchmarks also measure what the observed allocation costs. They price the deviation in dollars per campaign and count the readers the campaigns failed to inform.

5.1 The Advertiser’s Problem

A ministry maximizes the number of readers in the campaign’s stated demographic who see its advertisement at least once, subject to the campaign’s budget. Campaign j in year t carries budget B_{jt} . Outlet i charges p_{it} for one insertion and reaches a_{ijt} readers in that demographic, so spending s_{ijt} buys $x_{ijt} = s_{ijt}/p_{it}$ insertions.

Three assumptions turn that objective into a function of spending. A reader of outlet i sees any given insertion with probability ϕ , independently across insertions. Only the first exposure carries value and

readerships do not overlap across outlets. The first two fix what a reader of one outlet takes from a campaign, and the third lets me add readers across outlets.⁶ The ministry solves

$$\max_{\{s_{ijt} \geq 0\}} \sum_i a_{ijt} [1 - (1 - \phi)^{s_{ijt}/p_{it}}] \quad \text{s.t.} \quad \sum_i s_{ijt} \leq B_{jt} \quad (2)$$

The bracketed term gives the probability that a reader of outlet i sees the advertisement at least once, so the objective counts unique target readers. Reach then saturates within an outlet. A ministry that spends its whole budget at one newspaper reaches $a_{ijt}[1 - (1 - \phi)^{B_{jt}/p_{it}}]$ readers and never more than a_{ijt} , and the x -th insertion adds $\phi(1 - \phi)^{x-1}$ readers. The benchmark therefore spreads spending across outlets, and saturation accounts for most of the foregone reach reported below.

Federal campaigns transmit a fact, whether a social program has opened for applications, whether the clocks change on Sunday, or where a vaccine is available. A reader who already holds the fact gains nothing from a second insertion. Commercial advertising would violate the assumption, because repetition there builds persuasion and recall. The third assumption fails whenever a reader reads two newspapers, and equation 2 then counts that reader twice. The National Print Media Registry reports circulation and audience composition outlet by outlet and records nothing about who reads both, so I cannot measure the overlap. The benchmark buys more outlets per campaign than ministries do, so overlap would cut benchmark reach by more than observed reach, and the reach gap reported below is an upper bound. I calibrate ϕ below, and Appendix G reports the estimates across the range over which the budget binds.

5.2 Constructing the Benchmarks

Solving the problem reduces to finding one number per campaign. The objective is additively separable and strictly concave in each s_{ijt} and the constraint set is convex, so the Kuhn-Tucker conditions pin down a unique optimum. Marginal reach per dollar stays strictly positive everywhere, so the budget binds. Writing λ for its multiplier and $\kappa \equiv -\ln(1 - \phi)$,

$$s_{ijt}^*(\lambda) = p_{it} \cdot \max \left\{ 0, \frac{1}{\kappa} \ln \left(\frac{a_{ijt} \kappa}{\lambda p_{it}} \right) \right\} \quad (3)$$

The truncation selects which outlets a campaign funds. Outlet i returns $a_{ijt}\kappa/p_{it}$ readers on its first dollar, and every dollar after that returns less. An outlet whose first dollar already returns less than λ is worth nothing to the ministry at any scale, so its non-negativity constraint binds and the campaign buys none of it. The unconstrained first-order condition would report negative spending for those outlets, which the budget constraint would then hand back to the rest. Equation 3 therefore delivers an extensive margin as well as an intensive one, and Panel A of Table 4 compares the observed extensive margin against it.

⁶Gentzkow et al. (2024) endow each viewer with an intrinsic value to advertisers and let an advertiser who reaches a viewer M times obtain $a \cdot u(M)$ for a submodular $u(\cdot)$. Equation 2 uses the case $u(M) = \mathbf{1}[M > 0]$, which they call the reach-only model. Grossman and Shapiro (1984) write the reach of a campaign placed in m vehicles each covering a fraction r of the population as $1 - (1 - r)^m$ and label it the constant-reach, independent-readership technology.

I recover λ from the observed budget. Spending under equation 3 falls continuously and strictly as λ rises, so the budget fixes a unique value, which I locate by bisection campaign by campaign. The multiplier then translates into a reservation price. Write $c_{ijt} = p_{it}/a_{ijt}$ for what an outlet costs per target reader and set $\bar{c}_{jt} \equiv \kappa/\lambda$. A ministry buys outlet i if and only if $c_{ijt} < \bar{c}_{jt}$, and buys $x_{ijt}^* = \kappa^{-1} \ln(\bar{c}_{jt}/c_{ijt})$ insertions, so a campaign buys more from an outlet the cheaper that outlet is per target reader. The median reservation price reaches \$115 per thousand target readers, while outlets charge a median of \$89 with an interquartile range of \$58 to \$163. The reservation price therefore falls inside the price distribution, which is why some outlets enter the benchmark allocation and others do not. Appendix G reports how λ and $\bar{c}_{jt}$ vary across campaigns.

The exposure rate ϕ is not identified from the data, so rather than defend one value I report the estimates across its full range. Appendix G shows that the intensive-margin deviation stays positive and significant everywhere the budget constraint binds, so no conclusion turns on the calibration. I report the benchmark at $\phi = 0.10$, which places observed campaigns between two implausible extremes. A funded outlet receives two insertions at the median and five at the seventy-fifth percentile, which at $\phi = 0.10$ inform 19 and 41 percent of that outlet's readers. At $\phi = 0.02$ those two insertions would inform 4 percent, so a ministry buying print advertising would accomplish almost nothing. At $\phi = 0.25$ they would inform 44 percent, so a ministry would saturate an outlet after two insertions. Two quantities do move with the calibration. The benchmark funds one outlet at the concentration limit and 15.4 at the coverage limit, so Figure 4 supports a qualitative comparison rather than a point estimate.

Two audience weights answer two questions about what a ministry set out to buy. The *cost-efficiency benchmark* sets a_{ijt} to an outlet's readers in the campaign's stated demographic, which the legal cost-efficiency standard obliges ministries to buy.⁷ The *voter-targeting benchmark* sets a_{ijt} to the outlet's readers who support the incumbent, which I predict by weighting official state-level vote shares by where the outlet circulates. Strömberg (2004) shows that an office-seeking politician pursues exactly this objective. Appendix J checks the measure against an independent estimate built from individual polling microdata, with which it correlates at 0.80.

I estimate the same equation against each benchmark,

$$Y_{ijt} = \alpha \cdot \mathbf{1}[\text{Aligned}]_{it} + \gamma_j + \delta_i + \varepsilon_{ijt} \quad (4)$$

where $\mathbf{1}[\text{Aligned}]_{it}$ equals one when a Morena campaign meets a left-wing outlet or a PRI campaign meets a right-wing outlet. The outcome Y_{ijt} takes three forms. The extensive margin is $\mathbf{1}[s_{ijt}^{\text{obs}} > 0] - \mathbf{1}[s_{ijt}^* > 0]$ and the intensive margin is $s_{ijt}^{\text{obs}} - s_{ijt}^*$ in dollars, which I also divide by B_{jt} so that budgets ranging from thousands to millions of dollars weigh comparably. Campaign fixed effects absorb budget, stated demographic and

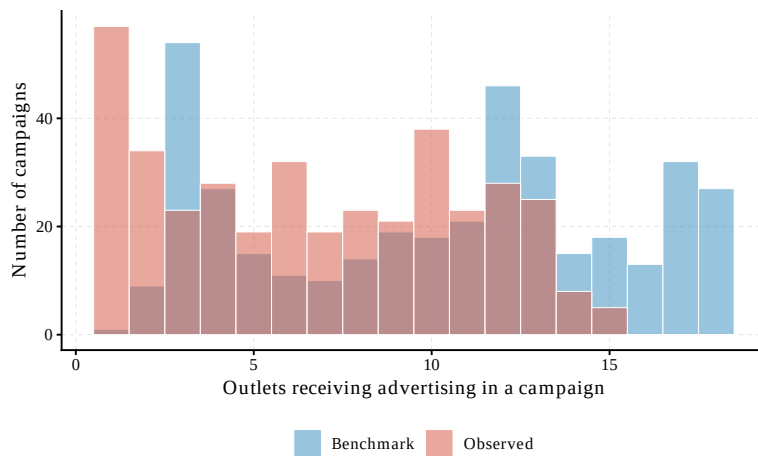
⁷Campaigns state a demographic on two dimensions, income and age, and the registry reports each outlet's readership shares on the same categories. I compute the share of an outlet's readers falling in the stated groups separately on each dimension and multiply the two, which treats income and age as independent within an outlet. The registry reports only marginal distributions, so no other combination is available. Three campaigns of 386 state a demographic that no national newspaper reaches, which leaves the benchmark undefined, so I drop them. The estimation sample holds 383 campaigns and 6,711 outlet-campaign cells.

party, and outlet fixed effects absorb every time-invariant outlet characteristic. I cluster standard errors by outlet.

5.3 Observed and Benchmark Allocations

Ministries buy fewer newspapers than reaching the stated demographic would require. The estimation sample covers 383 campaigns across 18 national outlets, giving 6,711 outlet–campaign cells. In Figure 4 I plot how many outlets a campaign funds, once under the allocation ministries chose and once under the benchmark. The observed distribution places its mode at a single outlet, so many campaigns run in one newspaper alone. Its median sits at six outlets and its mean at 6.7, against a median of eleven and a mean of 10.2 under the benchmark. The benchmark also places mass at fifteen outlets or more, where the observed distribution places almost none. Ministries therefore concentrate their campaigns beyond what informing the stated demographic would justify, and the within-campaign Herfindahl index of spending records the same gap at 0.36 against 0.19.

Figure 4: **Outlets Used per Campaign, Observed and Benchmark**



Outlets receiving positive advertising, across 383 campaigns. The benchmark solves equation 2 at $\phi = 0.10$ campaign by campaign, holding each observed budget fixed and letting equation 3 determine how many outlets receive funding. Bars overlap where the two distributions coincide.

The two administrations depart from the benchmark by different amounts. Reaching the benchmark would require moving 73.5 percent of a PRI campaign’s budget between outlets, and 81.4 percent of a Morena campaign’s, where that figure counts money changing outlet rather than money going to an outlet the benchmark rejects. Concentration follows the same order, since Morena campaigns fund 5.4 outlets against 7.3 under the PRI. I cannot compare the alignment gradient itself across administrations. Two parties and time-invariant outlet ideology make $\mathbf{1}[\text{Aligned}]$ equal to the product of a left-wing outlet indicator and a Morena indicator once the fixed effects absorb their main effects, so interacting alignment with the party in power reproduces alignment itself.

5.4 Favoritism in Allocation

Aligned outlets absorb the departure from either benchmark. In Table 4 I regress the gap between observed and benchmark spending on whether an outlet aligns with the party in power, holding campaign and outlet fixed, and I report each margin in its own panel. Panel A counts how often an outlet receives money the benchmark would withhold. Under the legal cost-efficiency standard aligned outlets receive such money 13.9 percentage points more often than misaligned outlets, or 11.4 points when the continuous ideology score replaces the binary split. Panel B prices the same gap at \$21,609 per campaign and \$28,897 respectively. Panel C divides each deviation by the campaign budget, which matters because budgets run from \$3,642 to \$7.7 million, and returns 2.82 and 3.49 percentage points. A benchmark campaign funds 10.2 outlets, so a funded outlet holds 9.8 percent of a budget on average and these estimates amount to roughly a third of that share. Columns (3) and (4) put the same question to a government maximizing reach to its own supporters, and every estimate rises rather than falls. Neither the ideological threshold nor the audience a ministry might have pursued changes the answer, because ministries directed money toward aligned outlets under every version.

Table 4: Deviations from the Benchmark Allocations by Outlet Alignment

	Cost-efficiency benchmark		Voter-targeting benchmark	
	(1)	(2)	(3)	(4)
<i>Panel A. Extensive margin</i>				
Alignment	0.139*** (0.032)	0.114*** (0.019)	0.144*** (0.038)	0.142*** (0.022)
<i>Panel B. Intensive margin, 2022 dollars</i>				
Alignment	21,609** (8,991)	28,897*** (5,651)	23,932** (9,808)	31,737*** (6,547)
<i>Panel C. Intensive margin, share of campaign budget (pp)</i>				
Alignment	2.82** (1.29)	3.49** (1.38)	3.40** (1.49)	4.11*** (1.41)
Alignment measure	Binary	Continuous	Binary	Continuous
Ad Campaign FE	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711

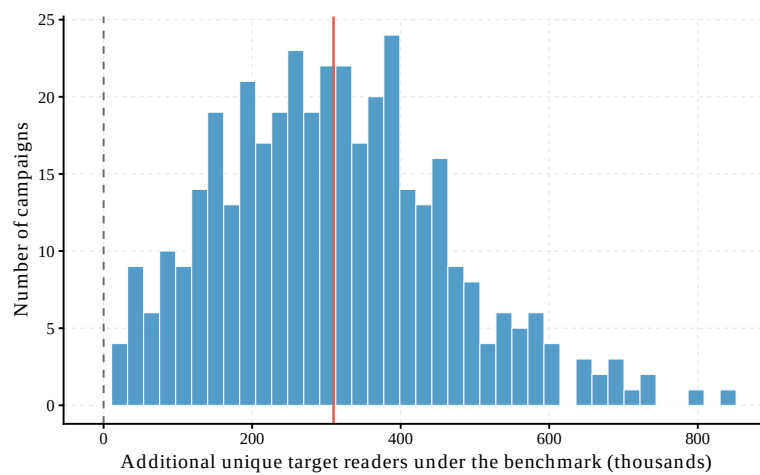
Estimates of equation 4 at $\phi = 0.10$. Panel A takes $1[\text{observed} > 0] - 1[\text{benchmark} > 0]$ as the outcome, Panel B takes the difference in 2022 dollars, and Panel C divides that difference by the campaign's total budget. Each column reports a separate regression in which the stated alignment measure is the only regressor. Binary columns split outlets at the ideological midpoint. Continuous columns replace the split with the ideology score of Section 3.2, signed so that higher values indicate closer alignment with the incumbent. Binary and continuous coefficients are measured on different scales and are not comparable in magnitude. Standard errors clustered by outlet across 18 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The copartisan measure weights state vote shares by where an outlet circulates, so it registers no sorting within a state and I cannot measure sorting directly. Appendix I instead rebuilds the voter-targeting benchmark under assumed sorting and reports how much of it the deviation survives. Measured as a share of the campaign budget, the deviation holds as long as the odds of an incumbent supporter reading an aligned rather than a misaligned newspaper stay within roughly 1.3 times what I measure, which would put 24 percent of an aligned outlet's readers on the incumbent's side against 21 percent of a misaligned

outlet's. Measured in dollars it holds to 2.7 times. Two further pieces of evidence make sorting beyond that hard to credit. Neither of the two independently constructed copartisan measures separates left-wing from right-wing outlets by even half a percentage point, as Appendix J reports. And Appendix F shows that every weighting of reader demographics predicts within-campaign allocation with the wrong sign, while a single parameter loading circulation on outlet ideology predicts it with the right one.

The benchmark also measures how many target readers the observed allocation forgoes. Campaigns inform between 178,000 and 275,000 target readers against the benchmark's 765,000, or 23 to 36 percent.⁸ Figure 5 plots that shortfall campaign by campaign, and the distribution lies entirely above zero, with a mean of 586,000 readers and a 95 percent confidence interval running from 550,000 to 622,000. Not one of the 383 campaigns informs as many readers as its own budget could have.

Figure 5: **Difference in Unique Target Readers Reached, Benchmark Minus Observed**



One observation per campaign. Reach counts readers in the campaign's named groups who see the advertisement at least once, computed at $\phi = 0.10$. Observed reach converts spending into insertions using the advertisement counts ministries recorded, and benchmark reach converts it at the private market price. The dashed line marks zero and the solid line marks the mean difference.

Outlet choice rather than price produces the shortfall. A ministry's own outlet choices, priced at the market rate, deliver 275,000 target readers, and the benchmark spends the same money to deliver 765,000. That 490,000 difference does not depend on the recorded advertisement counts, because both allocations convert spending into insertions at the same price. Appendix E decomposes the shortfall and finds that overpayment contributes almost nothing once the insertion count is reliable.

Ministries therefore departed from every allocation an audience objective would justify, and aligned outlets received what that departure moved. The deviation holds against the standard the law imposes and against the one an office-seeking politician would hold, it grows with an outlet's alignment, and no weighting of reader demographics reproduces it. Section 6 takes up what that revenue did once outlets received it.

⁸Ministries record both the number of advertisements they placed and the amount they paid, and the two do not always agree. The lower count uses the recorded advertisements. The higher converts spending into advertisements at the private market price, and it exceeds the first because ministries pay above that price and because the recorded counts appear to miss insertions in the cells where the discrepancy is largest, as Appendix E documents.

6 Downstream Effects in the Media Market

The previous sections documented systematic favoritism in government advertising allocation toward ideologically aligned outlets. These allocation patterns create asymmetric revenue shocks, with aligned outlets gaining resources while misaligned outlets lose a significant income stream. This section examines how these revenue shocks affect the structure of the media market and the production of news content.

6.1 Effects on Newsroom Capacity and Content Production

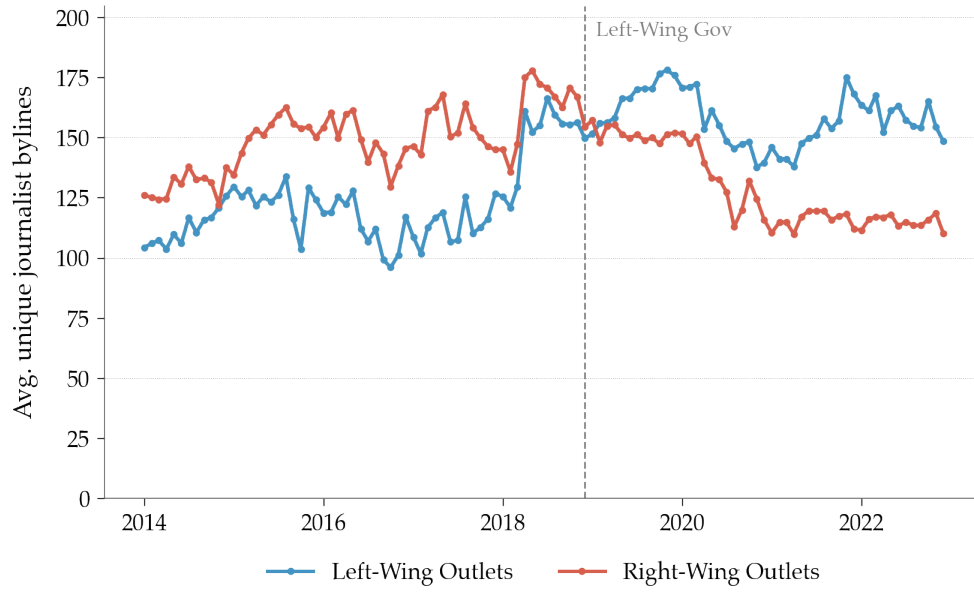
I assess whether shifts in government advertising following ideological transitions affect outlet operations and content production. If government advertising constitutes a substantial share of outlet revenue, then politically motivated reallocation affects newsroom resources — the size and composition of editorial staff — which in turn shapes the volume and quality of news production. Changes in content can then translate into shifts in readership demand. Using the same difference-in-differences design, I compare changes in newsroom size, content production, and circulation across aligned and misaligned outlets before and after the transition to a left-wing government. Finally, I examine whether the circulation gains are driven by partisan readers switching to ideologically aligned outlets or by readers attracted by the expansion in content production.

I begin by examining raw trends in newsroom size for left- and right-wing outlets around the 2018 government transition. Figure 6 presents these trends. Prior to the transition, right-wing outlets maintained larger newsrooms than left-wing outlets. After 2019, this pattern reverses, with left-wing outlets expanding substantially while right-wing outlets decline sharply, and left-wing outlets surpassing right-wing outlets for the remainder of the sample period. I argue that this divergence is partly driven by the reallocation of government advertising, which directly constrained right-wing outlets' capacity to maintain newsroom resources while enabling left-wing outlets to expand.

To formally test whether the differential trends in Figure 6 reflect the reallocation of government advertising, I estimate the difference-in-differences specification from equation 1, now using newsroom outcomes at the outlet-year level. Newsroom size is measured from byline analysis of the full article corpus, excluding agency content. A journalist is classified as active in a given month if they have published at least 16 articles across the entire study period. Appendix K.2 reports the detailed classification procedure and robustness checks. Table 5 presents the main results. Column (1) examines overall newsroom size. Columns (2) and (3) decompose by employment type, with Column (2) counting staff journalists (those with a primary or exclusive affiliation to the outlet) and Column (3) counting individual freelance journalists. The outcomes are transformed using the inverse hyperbolic sine; coefficients are interpretable as approximate percentage changes. All specifications include year and outlet fixed effects, with standard errors clustered at the outlet level.

Left-wing outlets experience substantial newsroom expansions following the transition. Column (1)

Figure 6: **Newsroom Size in Time across Right- and Left-Wing Outlets (2014–2022)**



Newsroom size represents the number of classified journalists active per outlet-month. A journalist is classified as active in a given month if they have published at least 16 articles across the entire study period. Journalists are identified through byline analysis, excluding agency content.

Table 5: Effect of Left-Wing Government on Newsroom Size

	<i>Dependent variable:</i>		
	IHS(Total Newsroom)	IHS(Staff)	IHS(Journalists)
	(1)	(2)	(3)
Left Gov × Left Outlet	0.600*** (0.133)	0.266*** (0.099)	0.844*** (0.206)
Mean pre-2018 (Left)	5.08	4.03	3.18
Mean pre-2018 (Right)	5.87	4.86	3.89
SD pre-2018	1.17	1.15	1.07
Year FE	✓	✓	✓
Outlet FE	✓	✓	✓
Observations	1,908	1,908	1,908
R ²	0.842	0.960	0.786

shows that total newsroom size increases by approximately 60% relative to right-wing outlets. This growth is driven by both employment types, with staff journalists expanding by approximately 27% (Column 2) and freelance journalists by approximately 84% (Column 3), indicating that aligned outlets both strengthen their permanent editorial staff and draw in a larger pool of contributing writers. Appendix K.2 reports the same results in levels and presents event study estimates confirming that the parallel trends assumption holds in the pre-transition period.

I then examine whether the newsroom expansion translates into increased content production. Table 6 presents the results of estimating the same difference-in-differences specification using the inverse hyperbolic sine of the monthly article count within each content type, so coefficients are interpretable as approximate percentage changes. Columns (1)–(4) cover hard news topics (total, politics, economics, and

social policy), and columns (5)–(6) cover soft news (sports and entertainment). All specifications include outlet and year fixed effects, with standard errors clustered at the outlet level.

Table 6: **Effect of Left-Wing Government on News Production by Content Type**

	<i>Dependent variable:</i>					
	Total	Politics	Economics	Social Policy	Sports	Entertainment
	(1)	(2)	(3)	(4)	(5)	(6)
Left-Wing Government × Left-Wing Outlet	0.416*** (0.154)	0.425*** (0.149)	0.512*** (0.196)	0.474** (0.189)	0.342 (0.235)	0.356* (0.213)
Year FE	✓	✓	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓	✓	✓
Mean pre-2018 (Left)	10.502	9.559	8.415	8.048	8.319	6.988
Mean pre-2018 (Right)	10.967	9.98	9.297	8.647	8.484	7.528
SD pre-2018	0.787	0.864	1.097	0.955	0.744	0.811
Observations	159	159	159	159	159	159
R ²	0.907	0.923	0.934	0.906	0.854	0.867

Note: *p<0.1; **p<0.05; ***p<0.01. SEs clustered at the outlet level. Outlet-month panel. The outcome is the inverse hyperbolic sine of the monthly count of articles within each content type, so coefficients are interpretable as approximate percentage changes. Cols. (1)–(4) cover hard news, and cols. (5)–(6) cover soft news. All columns include outlet and year fixed effects. Means and standard deviations are computed over the pre-2018 period in the units of the transformed outcome.

Left-wing outlets increase news production following the transition across all hard news categories. Total article production rises by approximately 42% (Column 1), corresponding to roughly 776 additional articles per month. The gains are largest in economics (approximately 51%) and social policy (approximately 47%), followed by politics (approximately 43%), consistent with a newsroom expansion concentrated in substantive, resource-intensive reporting. As a back-of-the-envelope check, the 776 additional monthly articles imply approximately 9 articles per additional journalist — somewhat below the pre-transition baseline of 13 — consistent with new hires ramping up over time. The effect on soft news is positive for entertainment (approximately 36%) and smaller and insignificant for sports, indicating that the extra resources translate primarily into hard news coverage rather than a broad undifferentiated expansion.

I next examine whether the content expansion reflects changes in article quality and sourcing composition. Column (1) of Table 7 reports the effect on the log of average article length among affiliated journalists; pre-transition means correspond to approximately 490 words for both outlet types. Columns (2) and (3) examine wire sourcing, measuring the IHS of the count of wire-sourced articles and the wire-sourced share of total articles, respectively. All specifications include outlet and year fixed effects with standard errors clustered at the outlet level.

Table 7: Effect of Left-Wing Government on Article Length and Wire Sourcing

	<i>Dependent variable:</i>		
	Log(Article Length)	IHS(Wire Articles)	Wire (%)
	(1)	(2)	(3)
Left-Wing Government $\times$ Left-Wing Outlet	-0.003 (0.056)	0.278 (0.401)	-0.416 (2.670)
Year FE	✓	✓	✓
Outlet FE	✓	✓	✓
Mean pre-2018 (Left)	6.189	5.537	12.611
Mean pre-2018 (Right)	6.164	6.394	15.185
SD pre-2018	0.168	1.516	7.118
Observations	1,899	1,908	1,908
R ²	0.722	0.637	0.533

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. SEs clustered at the outlet level. Col. (1) is the log of average article length among affiliated journalists, whose pre-transition means correspond to approximately 490 words for both outlet types. Col. (2) is the inverse hyperbolic sine of the count of wire-sourced articles and col. (3) is the wire-sourced share of total articles in percentage points. All columns include outlet and year fixed effects. Means and standard deviations are computed over the pre-2018 period in the units of the corresponding outcome.

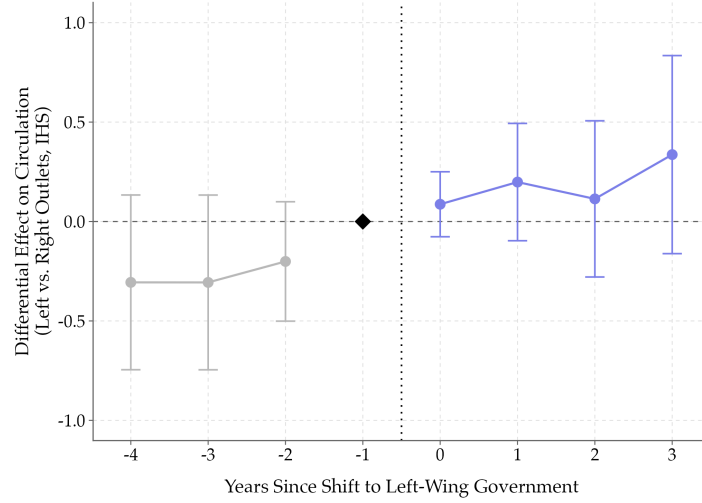
None of the three coefficients is statistically significant. Article length is unchanged (Column 1), indicating that affiliated journalists do not write longer or shorter articles on average. The count of wire-sourced articles rises modestly in absolute terms (Column 2), but the wire-sourced share of total articles is slightly negative (Column 3), suggesting that original reporting grows faster than agency-sourced content. Taken together, the content expansion documented above is driven by a larger number of articles of similar length and complexity, consistent with the newsroom growth translating into greater volume of original reporting rather than changes in quality per article.

6.2 Effects on Circulation

I now examine whether the gains in newsroom capacity and news production translate into readership growth through two distinct mechanisms. On the supply side, increased government advertising revenue directly enables outlets to hire more journalists and produce more content, which may attract readers through greater coverage. On the demand side, the 2018 election may have shifted reader preferences independently of any changes in content quality, as voters who moved toward Morena may also have increased their demand for ideologically aligned news.

To examine whether circulation changes following the 2018 transition, I estimate the event study specification from equation 1 using the inverse hyperbolic sine of circulation as the outcome. Because circulation is measured at the outlet-state level, the specification includes outlet-state and year fixed effects, with standard errors clustered at the outlet-state level. Figure 7 presents the estimates. Pre-transition coefficients are close to zero and statistically indistinguishable from zero, supporting the parallel trends assumption. Following the transition, left-wing outlets gain circulation relative to right-wing outlets, with the gap opening immediately after 2018 and reaching approximately 40 percent by the end of the sample period.

Figure 7: **Left-Wing Outlets Gain Circulation Relative to Right-Wing After 2018**



This figure presents event study estimates of the differential effect of the government transition on IHS(Circulation) across left- and right-wing outlets. Each point represents the interaction coefficient between event time and a binary indicator for left-wing outlet ideology. Event time -1 (the year before the December 2018 transition) is normalized to zero. Error bars represent 95% confidence intervals. The specification includes outlet-state and year fixed effects, with standard errors clustered at the outlet-state level.

To separate circulation gains driven by improvements in coverage from those driven by shifts in reader preferences, I exploit cross-state variation in the leftward shift in Morena's vote share between 2015 and 2018. The intuition is that content improvements at left-wing outlets are national — the same newspaper is distributed across all Mexican states — while preference-driven gains should be larger where more voters shifted toward Morena. States with little or no leftward vote shift therefore serve as a counterfactual for the supply-side component; with preferences largely unchanged in those states, any differential circulation gains for left-wing outlets reflect content improvements alone. The additional gains in high-shift states then identify the preference-driven component. This design relies on two assumptions. First, content effects are uniform across states. Second, demand-side shifts are proportional to the local electoral shift. In particular, I estimate a triple difference-in-differences specification:

$$\begin{aligned}
 Y_{nst} = & \beta_1 (\text{Left-Wing Gov}_t \times \text{Left-Wing Outlet}_n) \\
 & + \beta_2 (\text{Left-Wing Gov}_t \times \text{Left-Wing Outlet}_n \times D_s) \\
 & + \beta_3 (\text{Left-Wing Gov}_t \times D_s) + \alpha_{ns} + \gamma_t + \varepsilon_{nst}
 \end{aligned} \tag{5}$$

where $D_s = 1[\Delta \text{Vote Share Left-Wing Party}_s > \text{median}]$ is an indicator for states with an above-median increase in left-wing vote share between 2015 and 2018, α_{ns} are outlet-state fixed effects, and γ_t are year fixed effects. Appendix Figure K3 maps the state-level variation in Morena's vote share shift that defines D_s ; southern and southeastern states experienced shifts of 50–65 percentage points, while northern states shifted by 20–35 percentage points. Under this design, β_1 captures the content-driven component of circulation gains, common across all states regardless of political shift. β_2 identifies the additional preference-driven component, and β_3 tests whether the overall political shift drives circulation independently of outlet

ideology.

Table 8 presents the results. The coefficient on Left-Wing Gov $\times$ Left-Wing Outlet is 0.397, statistically significant at the 10% level. The interaction with the high-shift indicator (Left-Wing Gov $\times$ Left-Wing Outlet $\times$ D_s) is 0.173, positive but not statistically significant. The coefficient on Left-Wing Gov $\times$ D_s alone is -0.003 , indicating that the overall leftward political shift does not independently drive circulation across all outlet types. Weighting $\hat{\beta}_2$ by the share of high-shift states — one half, by construction of the median split — the content-driven component ($\hat{\beta}_1 = 0.311$) accounts for approximately 78% of the total average circulation gain, equivalent to roughly a 31% increase in circulation attributable to content improvements alone. The remaining 22% reflects preference shifts concentrated in states with above-median electoral realignment toward Morena.

Table 8: **Decomposing Circulation Gains**

	(1) Baseline DiD	(2) Triple DiD
Left-Wing Gov $\times$ Left-Wing Outlet	0.397* (0.225)	0.311* (0.178)
Left-Wing Gov $\times$ Left-Wing Outlet $\times$ D_s		0.173 (0.124)
Left-Wing Gov $\times$ D_s		-0.003 (0.064)
Outlet $\times$ State FE	✓	✓
Year FE	✓	✓
Sample	All	All
Observations	5,194	5,194
R ²	0.957	0.957

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Outcome: $\text{IHS}(\text{Circulation})$. Clustered standard errors at the outlet $\times$ state level. $D_s = 1[\Delta \text{Vote Share Left-Wing Party}_s > \text{median}]$.

Together, the event study and decomposition results establish that the circulation gains documented at left-wing outlets following the 2018 transition are real, persistent, and driven predominantly by improvements in content rather than by shifts in reader preferences. This pattern completes the causal chain from government advertising to media market outcomes — resource transfers enable newsroom expansion, expansion generates greater news production, and the resulting content improvements attract readers independently of any change in audience ideology.

7 Conclusion

This paper demonstrates that governments systematically use advertising allocations to favor ideologically aligned media outlets. In places where government advertising represents a critical revenue source for news outlets, reallocations become powerful tools for reshaping media markets. I find that aligned outlets expand newsroom capacity and increase news production, while misaligned outlets contract substantially. I further show that observed allocations exceed what cost-efficiency or voter-targeting objectives would justify, and that circulation gains at aligned outlets are driven primarily by content improvements rather

than shifts in reader preferences.

These findings represent a novel mechanism of media capture in competitive markets. Rather than directly controlling content or suppressing critical voices through censorship or quid pro quo exchanges, governments can selectively subsidize aligned outlets, enabling them to maintain or expand operations while misaligned outlets decline. The resulting asymmetry in production capacity alters the information environment — aligned outlets gain resources to produce more news, while misaligned outlets reduce coverage. In competitive media markets where outlets depend on advertising revenue for survival, strategic allocation of government advertising becomes an effective instrument for influencing the structure of public discourse without overt interference in editorial independence.

The mechanism documented here has implications for democratic accountability. By selectively expanding the production capacity of ideologically aligned outlets while constraining misaligned ones, governments can reshape the information environment available to voters without overt censorship or direct editorial control — potentially limiting citizens' access to diverse perspectives and their ability to hold governments accountable for policy failures.

The mechanism identified here is most relevant in competitive democracies with active media markets — settings in which outlets compete for readers while depending on government advertising as a meaningful source of revenue. These conditions are common across Latin America, Eastern Europe, and other contexts where private advertising markets are thin and public clients are large. In such environments, the discretionary allocation of government advertising may be among the most consequential and least visible instruments of political influence over the press.

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A Setting and Data Sources

A.1 Regulatory Framework for Government Advertising in Mexico

Constitutional foundation (2007)

The modern regulation of official advertising in Mexico originates in the reform to Article 134 of the *Constitución Política de los Estados Unidos Mexicanos* enacted on 13 November 2007. The reform added paragraphs seven and eight, establishing two obligations. First, all public servants must administer public resources impartially and refrain from using their position to influence electoral competition. Second, any government propaganda—under any modality of social communication, at any level of government—must be strictly institutional in character, with informational, educational, or social-orientation purposes only; in no case may such propaganda include the names, images, voices, or symbols of any public servant. The 2014 political-electoral reform added further language to Article 134 and explicitly mandated Congress to pass a general regulatory law. Congress failed to do so for four years, prompting the *Suprema Corte de Justicia de la Nación* (SCJN) to declare the omission unconstitutional in November 2017 and set 30 April 2018 as a hard deadline.

The pre-2018 administrative framework

Between 2007 and 2018, the constitutional prohibition was operationalized not through statute but through a patchwork of administrative instruments:

1. **Annual *Lineamientos* (Acuerdos) issued by Segob.** Every December, the Secretaría de Gobernación published an *Acuerdo* in the DOF titled “*Lineamientos generales para las campañas de comunicación social de las dependencias y entidades de la Administración Pública Federal para el ejercicio fiscal [year]*”. These documents—traceable at least to fiscal year 2003–2004—already required each federal entity to submit an *Estrategia Anual de Comunicación Social* and a *Programa Anual de Comunicación Social* to Segob in January, to obtain prior authorization for every campaign, to use official airtime (*Tiempos Oficiales*) before purchasing commercial spots, and to report all advertising expenditures. A version of the national media registry (*Padrón de Medios*) and the COMSOC spending-disclosure system were already operational under these annual rules.
2. **Annual *Presupuesto de Egresos de la Federación* (PEF).** Each year’s budget decree included a dedicated article (e.g., Art. 17 of the PEF 2015) setting spending caps and supplementary rules for government advertising, providing a second layer of annual controls.
3. **General legislation.** The *Ley Orgánica de la Administración Pública Federal* (Art. 27, frac. XXXIX) delegated authority over official communications to Segob. The *Ley Federal de Presupuesto y Responsabilidad Hacendaria* (Art. 38), the *Ley Federal de Telecomunicaciones y Radiodifusión*, the *Ley General de Instituciones*

y Procedimientos Electorales, and the *Ley Federal de Transparencia y Acceso a la Información Pública* collectively provided the statutory basis on which the annual *Lineamientos* rested. The 2015 *Lineamientos* alone cite more than fifteen legal instruments as their foundation.

This administrative framework was, however, limited to federal entities. State and municipal governments were not formally bound by the *Lineamientos*.

The *Ley General de Comunicación Social* (2018)

The law was published in the DOF on 11 May 2018 (under President Enrique Peña Nieto) and entered into force on 1 January 2019. Its main substantive provisions are:

- **Scope extended to all levels of government** (Art. 3). For the first time, the planning, authorization, registry, and transparency requirements applied to federal, state, and municipal entities alike.
- **Annual planning requirement** (Arts. 22–25). Every public entity with an advertising budget must submit an *Estrategia Anual* and *Programa Anual de Comunicación Social* to the corresponding *Secretaría Administradora* (Segob at the federal level) in the first fortnight of January.
- **National Media Registry** (Arts. 35–37). Media outlets must register in the *Padrón Nacional de Medios de Comunicación*, administered by Segob, to be eligible for government advertising contracts. The registry is public.
- **Priority for official airtime** (Arts. 15–18, 26.VI). Federal entities must use *Tiempos Oficiales* (free state and fiscal airtime on radio and television) before purchasing *Tiempos Comerciales*.
- **Prohibition of personalized promotion** (Art. 9.I). Campaigns may not highlight, in a personalized manner, the names, images, voices, or symbols of any public servant (restating the 2007 constitutional rule).
- **Electoral blackout** (Art. 21). All government campaigns must be suspended during federal and local electoral campaign periods, with narrow exceptions for electoral authorities, health and education campaigns, civil-protection emergencies, and campaigns specifically authorized by the INE.
- **Mandatory transparency reporting** (Arts. 40–43). Entities must maintain updated spending data on their transparency portals (by medium, provider, contract, and campaign); submit a semi-annual report; and the *Secretaría Administradora* must report bimestrally to the relevant legislative chamber.
- **Social program disclaimer** (Art. 11). Any campaign promoting a subsidy or benefit program must display the legend: “*Este programa es público, ajeno a cualquier partido político. Queda prohibido el uso para fines distintos a los establecidos en el programa*”.

Critics immediately labeled the law the “*Ley Chayote*”, arguing that it legitimized rather than reformed the existing system. The Inter-American Commission on Human Rights concluded that the law “regularizes and generalizes bad practices” and failed to meet international standards on official advertising, noting the absence of objective distribution criteria, an independent oversight body, and meaningful sanctions.⁹

Subsequent reforms (2022)

The law has been amended twice within the 2014–2022 window relevant to this paper:

- **Reform 01** (DOF 12 April 2022): Added subsection VI Bis to Article 8, requiring government campaigns to use inclusive language free of discrimination and gender-based violence.
- **Reform 02** (DOF 27 December 2022): Substantive reform to multiple provisions, modifying the objectives of the law to emphasize the right of citizens to information and government accountability, and amending spending-limit rules for state-level entities.

Summary of versions

Version	DOF date	Key change
Original decree	11 May 2018	First statutory law; in force 1 Jan 2019
Reform 01	12 April 2022	Art. 8 VI Bis: inclusive-language requirement
Reform 02	27 December 2022	Multiple provisions; accountability framing

Table A1: Enacted versions of the regulatory law within the 2014–2022 period.

All versions cited above are available in [source/1gcs/](#) of the project repository.

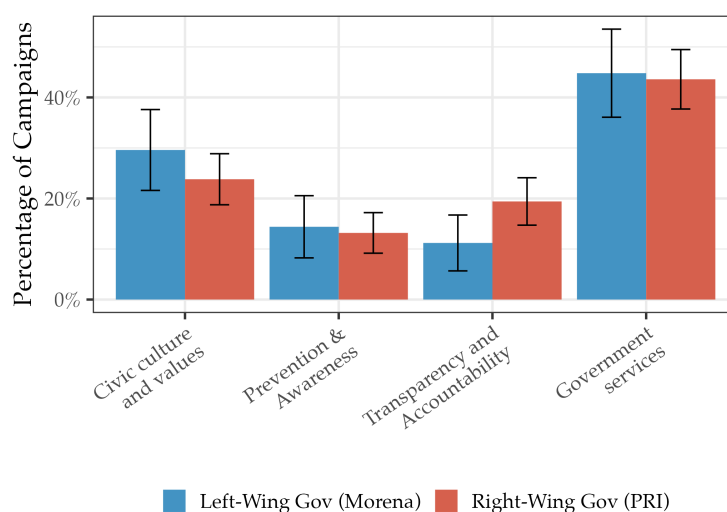
Table A2: Government Advertising Budgets by Ministry and Party (Millions MXN)

Ministry	PRI		Morena	
	Millions	%	Millions	%
Health	361.82	21.0	366.14	46.0
Finance	235.09	14.0	90.99	11.0
Education	228.66	14.0	1.36	0.0
Energy	121.22	7.0	91.60	11.0
Social Development	180.11	11.0	29.30	4.0
Environment	141.00	8.0	48.56	6.0
Communications and Transportation	104.98	6.0	21.91	3.0
Labor	83.58	5.0	15.26	2.0
Interior	35.96	2.0	31.21	4.0
Navy	34.32	2.0	30.68	4.0
Agrarian, Territorial and Urban Development	59.37	4.0	0.00	0.0
National Defense	10.85	1.0	37.36	5.0
Foreign Affairs	29.87	2.0	5.08	1.0
Presidency	20.41	1.0	8.07	1.0
Culture	12.28	1.0	11.22	1.0
Economy	15.18	1.0	0.00	0.0
Agriculture	7.50	0.0	6.16	1.0
Tourism	8.65	1.0	1.15	0.0
Public Administration	0.00	0.0	1.07	0.0

The table shows government advertising expenditures during the two administrations across federal ministries. Total spending is reported in millions of Mexican pesos, and the percentage represents the proportion of each ministry's budget for a given administration.

A.2 Government Advertising Descriptives

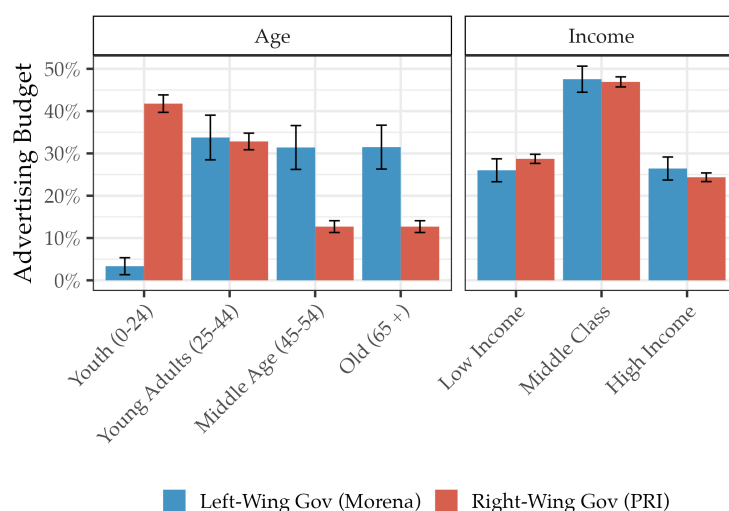
Figure A1: Share of Advertising Campaigns by Content Type and Party



The figure shows the distribution of government advertising campaigns across campaign types, as reported by the contracting government ministry. Campaigns are classified by the party in power at the time of placement. The sample includes 273 campaigns during the PRI administration and 125 campaigns during the Morena administration, covering the period 2014–2022.

⁹IACHR, *Principles on the Regulation of Official Advertising and Freedom of Expression* (2018 assessment).

Figure A2: Share of Advertising Campaigns Targeting Each Demographic Group by Party



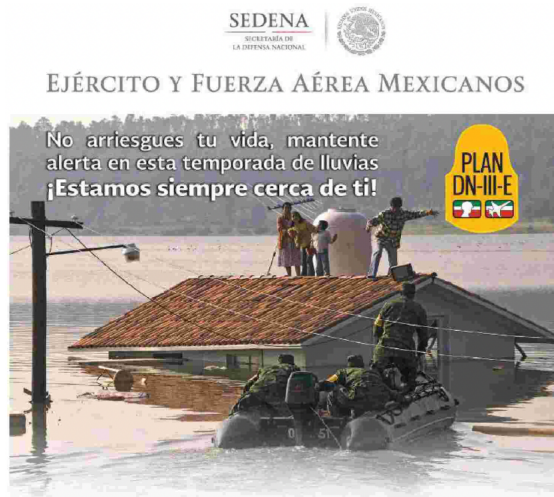
The figure shows the demographic groups targeted by government advertising campaigns conditional on the campaign specifying a target audience, as reported by the contracting government entity. Campaigns are classified by the party in power at the time of placement. Campaigns that report no specific targeting are excluded. The full sample consists of 398 government communication campaigns between 2014 and 2022.

Table A3: Bipartisan Government Advertising Campaigns

Ministry	Type	Title	Total Spending (USD)	
			PRI	Morena
Health	Prevention	Obesity Prevention	1,946,087	2,786,190
Interior	Civic Culture	Human Rights	66,200	1,895,410
Navy	Government Services	National Maritime Authority	1,325,999	157,456
Navy	Civic Culture	The Navy Close to You	1,181,894	131,892
Defense	Civic Culture	Military Commemorations	891,358	276,259
Foreign Affairs	Government Services	Protection of Mexicans Abroad	1,089,583	35,021
Education	Civic Culture	Sports in Mexico	1,087,621	25,974
Defense	Civic Culture	Commitment to Mexico	432,652	379,743
Defense	Civic Culture	Army and Family Outreach	384,237	58,197
Presidency	Civic Culture	In Search of Equality	153,061	209,533
Tourism	Prevention	Green Angels Road Service	302,958	11,356
Culture	Civic Culture	Linguistic Diversity	164,744	145,441
Defense	Civic Culture	Military Education	214,854	44,258
Tourism	Transparency	Local Tourism Promotion	214,352	23,920
Energy	Civic Culture	Daylight Saving Time	109,721	4,525
Education	Civic Culture	National Sports Award	2,710	18,701

The table shows government advertising campaigns that did not change their target audience during the sample period. Spending during the PRI administration covers 2014 to November 30, 2018; spending during the Morena administration covers December 1, 2018 to 2022.

Figure A3: Example of Cross-Administration Campaign: Ministry of Defense



(a) PRI Administration (2014–2018)



(b) Morena Administration (2018–2022)

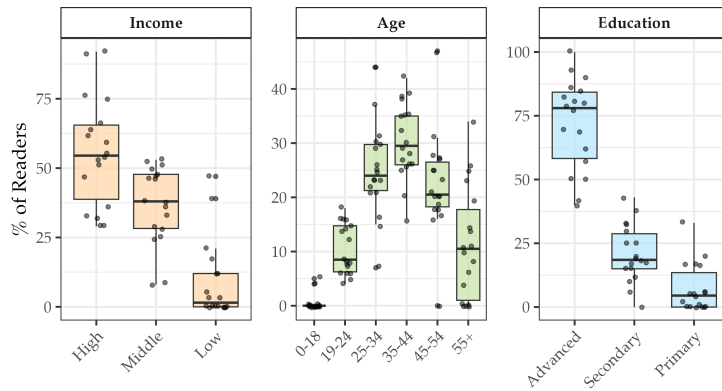
A.3 Media Market

Table A4: Summary Statistics: News Articles

Newspaper	N Total Articles	Articles per Year	
		Mean	SD
La Prensa	198,133	22,014.78	5,148.32
Esto	129,351	14,372.33	8,254.49
Ovaciones	245,451	27,272.33	7,839.42
Reporte Indigo	81,175	9,019.44	1,916.97
Publimetro	72,942	8,104.67	2,795.21
El Universal	399,328	44,369.78	7,629.34
Reforma	622,705	69,189.44	9,873.62
La Jornada	397,990	44,221.11	3,524.11
Excelsior	545,266	60,585.11	9,675.98
Milenio	339,008	37,667.56	8,352.47
La Razon	59,330	17,703.33	2,460.09
El Financiero	203,748	22,638.67	5,182.94
24 Horas	157,708	17,523.11	2,282.09
La Crónica de Hoy	253,724	28,191.56	8,771.31
El Sol de México	373,994	41,554.89	13,712.01
El Heraldo	143,123	17,890.38	11,760.17
Diario de México	83,352	9,261.33	3,337.80
El Economista	261,679	29,075.44	3,260.44

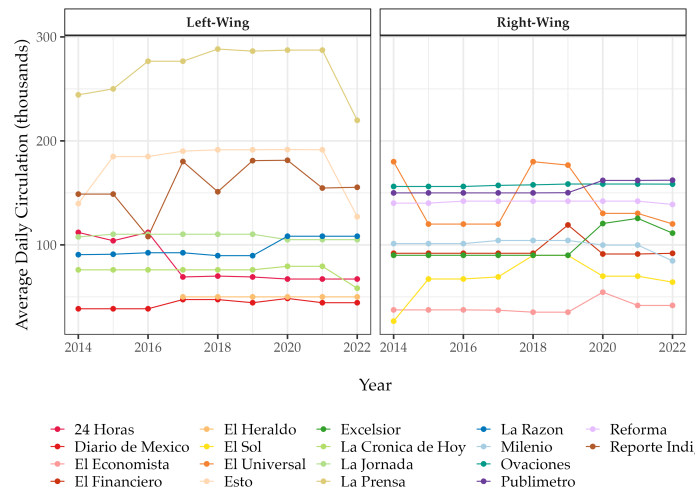
This table reports summary statistics of news articles for 18 national newspapers from 2014 to 2022. N Total Articles shows the total number of articles published by each outlet over the entire period. Articles per Year Mean and SD report the average number of articles per year and the corresponding standard deviation.

Figure A4: Reader Demographics Distribution Across Newspaper Outlets (2022)



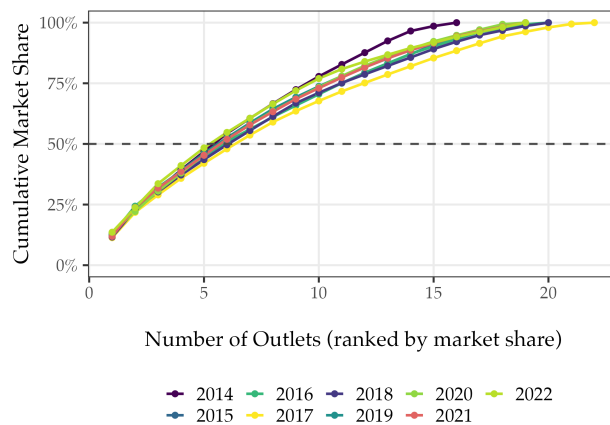
This figure displays the distribution of reader demographics across 18 newspaper outlets in 2022. Each panel shows boxplots representing the variation in the share of readers within each demographic category. Income levels are classified as High (AB/C+), Middle (C/D+), and Low (D/E). Education categories include Advanced, Secondary, and Primary education.

Figure A5: Circulation by Outlet (2014–2022)



This figure displays the average daily circulation for each outlet from 2014 to 2022. Data comes from the National Electoral Institute through their media monitoring program.

Figure A6: Cumulative Market Share (2014–2022)



This figure shows the cumulative distribution of market share across the universe of national print media outlets for each year from 2014 to 2022. Outlets are ranked within each year by their circulation, from largest to smallest.

Table A5: Ownership Structure of National Newspapers

Newspaper	In sample	Owner	Conglomerate	Market share	
				Mean	SD
El Grafico		Ealy Ortíz	Grupo Universal	12.30	0.75
La Prensa	1	Vázquez Raña	OEM	11.50	1.13
Esto	1	Vázquez Raña	OEM	7.57	0.80
Ovaciones	1	Vázquez Raña	OEM	6.81	0.75
Reporte Indigo	1	Garza García	Mac Multimedia	6.69	1.29
Publimetro	1	Rivas Garza	Publicaciones Metropolitanas	6.66	0.83
El Universal	1	Ealy Ortíz	Grupo Universal	6.16	1.59
Reforma	1	Junco de la Vega	Grupo Reforma	6.09	0.66
La Jornada	1	–	Demos	5.23	1.81
Mas por Mas		–	–	4.85	2.52
Uno Mas Uno		Libien Kauí	–	4.77	1.35
Excelsior	1	Vázquez Aldir	Grupo Imagen Multimedia	4.34	1.03
Milenio	1	González Family	Multimedios	4.32	0.47
La Razon	1	Garza Cantú	LRHG Informativo	4.19	0.65
El Financiero	1	Arroyo Rodríguez	Grupo Lauman	3.83	0.89
24 Horas	1	Zabludovsky	Información Integral 24/7	3.67	1.13
La Crónica de Hoy	1	Kahwagi Gastine	La Crónica	3.16	0.50
El Sol de México	1	Vázquez Raña	OEM	2.91	0.63
El Heraldo	1	Mieres Zimmermann	Grupo Andrade	2.09	0.18
Diario de México	1	–	Editorial DDM	1.85	0.21
El Economista	1	Nacer Gobera	–	1.73	0.37
Eje Central		Oliver Fernández	OFEM Media Group	0.63	0.09

Market share is computed as the outlet's share of total national print circulation in a given year, averaged over 2014–2022. Ownership and conglomerate information is obtained from the National Registry of Print Media (PNMI) and the Media Ownership Monitor Mexico database. When ownership information is not reported, this reflects either the absence of a clearly identifiable controlling shareholder or a cooperative ownership structure.

Table A6: Classification of Journalists in Database (2014–2022)

Employment Pattern	Category	N	%
Exclusive Affiliation	<i>Total</i>	2,560	43.10
	Left-Wing Outlets	1,163	12.97
	Right-Wing Outlets	1,397	15.58
Primary Affiliation	<i>Total</i>	2,214	37.27
	Left-Wing Outlets	869	9.69
	Right-Wing Outlets	1,345	15.00
Outlet Movers	<i>Total</i>	380	6.39
	Within-Ideology Moves	214	2.38
	Left → Right	78	0.87
	Right → Left	75	0.84
	Multiple Switches	14	0.16
Freelance/Multi-Outlet		785	13.21
Total		5,939	100.00

This table summarizes the classification of 5,939 journalists in the database. *Exclusive Affiliation* journalists publish exclusively at one outlet. *Primary Affiliation* journalists publish predominantly at one outlet but occasionally contribute to others. *Outlet Movers* switched between outlets during the period, either within the same ideological category or across ideologies. *Freelance/Multi-Outlet* exhibit diverse publication patterns across multiple outlets.

B Robustness: Alternative Ideology Measures

Table B1 replicates the main analysis using the continuous ideology score rather than the binary classification. The three outcomes are the same as in Table 3. The interaction term is Left-Wing Gov $\times$ Outlet's Ideal Score, where higher scores indicate more right-leaning positions. A negative coefficient indicates that more right-wing outlets receive less advertising under the left-wing government, consistent with ideological favoritism.

Table B1: Robustness: Continuous Ideology Score

	All Campaigns			Bipartisan Campaigns	
	Prob(Adv > 0) (1)	IHS(Adv) (2)	Price Premium (3)	Prob(Adv > 0) (4)	IHS(Adv) (5)
Left-Wing Gov $\times$ Outlet's Ideal Score	-0.161* (0.061)	-2.193* (0.838)	-0.212*** (0.027)	-0.116* (0.049)	-1.357** (0.474)
Mean (Pre-2018, Left)	0.46	—	—	0.26	—
Mean (Pre-2018, Right)	0.66	—	—	0.36	—
Year FE	✓	✓	✓	✓	✓
Outlet $\times$ Ministry FE	✓	✓		✓	✓
Campaign FE			✓		
Observations	2,880	2,880	2,617	1,440	1,440

All columns use the full outlet–ministry–year panel across all ministries, so the number of observations exceeds that in Table 3, which restricts to the ten ministries running bipartisan campaigns. Outlet's Ideal Score is the continuous GST score; higher values indicate more right-leaning positions. A negative coefficient indicates that more right-wing outlets receive less advertising under the left-wing government. SEs clustered at the newspaper level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B2 replicates the analysis using a three-category ideology classification. Outlets are divided into terciles of the continuous score (6 outlets each): left-wing, centrist, and right-wing. The reference category is right-wing outlets. Both left-wing and centrist outlets gain advertising relative to right-wing outlets under the left-wing government, with left-wing outlets gaining the most.

Table B2: Robustness: Three-Category Ideology Classification

	All Campaigns			Bipartisan Campaigns	
	Prob(Adv > 0) (1)	IHS(Adv) (2)	Price Premium (3)	Prob(Adv > 0) (4)	IHS(Adv) (5)
Left-Wing Gov $\times$ Left-Wing Outlet	0.288*** (0.063)	3.890*** (0.774)	0.283** (0.089)	0.171* (0.066)	1.924* (0.688)
Left-Wing Gov $\times$ Centrist Outlet	0.229** (0.060)	3.119*** (0.764)	0.008 (0.139)	0.133* (0.062)	1.457* (0.670)
Mean (Pre-2018, Left)	0.46	—	—	0.26	—
Mean (Pre-2018, Right)	0.66	—	—	0.36	—
Year FE	✓	✓	✓	✓	✓
Outlet $\times$ Ministry FE	✓	✓		✓	✓
Campaign FE			✓		
Observations	2,880	2,880	2,617	1,440	1,440

Reference category: right-wing outlets (6 most right-wing by GST score). Left, centrist, and right-wing categories are terciles of 18 outlets ranked by continuous GST score (6 outlets each). SEs clustered at the newspaper level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B.1 Outlets Scored Without Editorials

Diario de México, El Heraldo de México, and Esto publish no editorial section and are scored by applying the estimated phrase loadings to their news articles, as described in Section 3.2. Two features of these corpora differ from the editorial corpora used for the other fifteen outlets. They are far larger, with 26,493 articles for Esto against a median of 469 editorials per outlet, and they contain agency copy, which carries no editorial stance. Table B3 addresses both. I first restrict each corpus to journalist-authored content, dropping wire and agency material, and then draw 500 bootstrap samples at the median and mean editorial corpus size, rescoreing each draw.

The journalist restriction moves each score by less than 0.02, and the bootstrap confirms that corpus size is not what drives the classification of Diario de México or Esto, whose signs are stable in 99.8 and 100 percent of draws respectively. El Heraldo de México is genuinely ambiguous. Its score sits within 0.03 of the anchor midpoint and its sign is stable in only 78 to 83 percent of draws. It is coded as left-wing in the main analysis. The outlet enters a single ministry–outlet series and the main estimates are unchanged when it is coded as right-wing.

Table B3: Bootstrap Validation for Outlets Scored From News Articles

	Articles (all)	Journalist articles	Score (all)	Score (journalist)	95% CI (bootstrap)	Sign stable
Diario de México	5,102	4,539	+0.114	+0.102	[+0.025, +0.174]	99.8%
El Heraldo de México	3,081	2,903	−0.034	−0.027	[−0.097, +0.051]	77.8%
Esto	26,493	21,502	−0.296	−0.296	[−0.347, −0.245]	100.0%

Scores are recentered at the anchor midpoint, so negative values indicate left-leaning language. Journalist articles exclude wire and agency copy. Bootstrap draws 500 samples of 469 articles, the median editorial corpus size, from the journalist-authored corpus and rescores each draw. Sign stable reports the share of draws whose sign matches the full-corpus classification. At the mean editorial corpus size of 596 articles the corresponding shares are 99.8, 82.6, and 100.0 percent.

C Empirical Evidence: Favoritism in Allocation

C.1 Robustness

Section 4 reports the intensive margin in levels rather than under a log-like transformation. Table C1 reports both side by side, since the prior literature estimates log-type specifications and a reader coming from that work needs the comparable coefficient. The specification is otherwise identical to Table 3, so columns (1) and (3) reproduce its columns (2) and (5) exactly.

The inverse hyperbolic sine estimates are positive and statistically significant in both panels. They should not, however, be read as percentage changes. Because a substantial share of outlet–ministry pairs receive no advertising in a given year, the transformation is evaluated in a region where it does not approximate the logarithm, and the coefficient depends on the units in which spending is denominated. Table C2 makes that dependence explicit. Rescaling the outcome by a factor of one hundred moves the estimate by 36 to 41 percent. This is why the level effect is the one reported in the body (Chen and Roth, 2024), and why the

semi-elasticity in Table C3, which is estimated by Poisson and is invariant to scale, is the appropriate object if a proportional magnitude is wanted.

Table C1: Intensive Margin in Levels and Under the Inverse Hyperbolic Sine

	All Campaigns		Bipartisan Campaigns	
	Level (2022 USD) (1)	asinh (2)	Level (2022 USD) (3)	asinh (4)
Left-Wing Gov × Left-Wing Outlet	105,362* (53,615)	2.365** (1.019)	15,913* (9,017)	1.316** (0.592)
Mean (pre-2018, left)	107,562	6.49	7,563	2.66
Mean (pre-2018, right)	160,559	9.05	16,484	3.83
Year FE	✓	✓	✓	✓
Outlet × Ministry FE	✓	✓	✓	✓
Observations	1,410	1,410	1,410	1,410

Columns (1) and (3) report advertising spending in 2022 USD and reproduce columns (2) and (5) of Table 3. Columns (2) and (4) replace the outcome with its inverse hyperbolic sine. All columns use the outlet–ministry–year panel restricted to the ten ministries that ran bipartisan campaigns, with year and outlet–ministry fixed effects and standard errors clustered at the outlet level. Pre-2018 means are reported in the units of the corresponding outcome. The inverse hyperbolic sine coefficients are not interpretable as percentage changes and are sensitive to the scaling of the outcome; see Table C2. *p<0.1; **p<0.05; ***p<0.01.

Table C2: Robustness to Outcome Scaling

Specification	Treatment Effect Using:			Change from Rescaling Units:	
	arcsinh(Y)	arcsinh(100×Y)	Extensive Margin	Raw	%
Ministry-Outlet (1)	2.365	3.205	0.183	0.841	35.5
Bipartisan Campaigns (2)	1.316	1.854	0.117	0.538	40.9
Campaign-Outlet Level (3)	1.213	1.745	0.116	0.532	43.9

This table examines the sensitivity of treatment effect estimates to different transformations of the outcome variable. Column 2 reports the coefficient from a regression using $\text{arcsinh}(Y)$ as the dependent variable. Column 3 reports the coefficient using $\text{arcsinh}(100 \times Y)$. Column 4 shows the treatment effect on the extensive margin (whether $Y > 0$).

C.2 Intensive Margin Robustness: Poisson and Lee Bounds

Following Chen and Roth (2024), Table C3 reports two alternative estimators for the intensive margin. Column (1) presents a Poisson PPML regression, which is scale-invariant; the coefficient can be interpreted as $\exp(\hat{\beta}) - 1 \approx 222\%$, representing the proportional increase in expected spending for aligned outlets. Column (2) reports Lee bounds (Lee, 2009) on the conditional intensive margin effect — the effect for ministry–outlet pairs that receive positive advertising under both administrations — which range from \$11,668 to \$373,764. Both exercises confirm a large positive effect on the intensive margin.

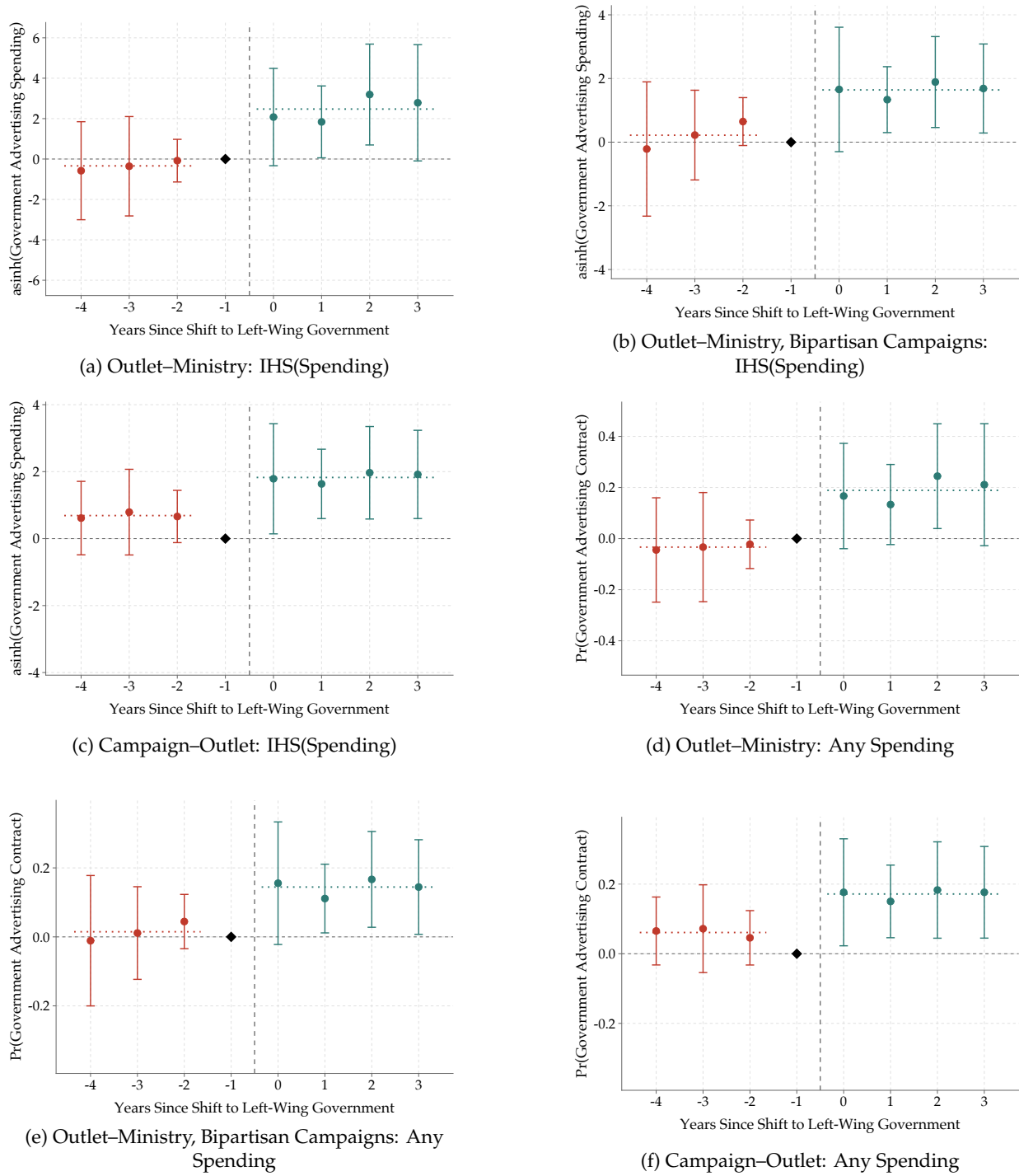
Table C3: Intensive Margin Robustness: Poisson and Lee Bounds

	Poisson PPML (semi-elasticity) (1)	Lee Bounds (2022 USD, conditional intensive margin) (2)
Left-Wing Gov × Left-Wing Outlet	1.169** (0.452)	[+11,668 ; +373,764]
Year FE	✓	
Outlet × Ministry FE	✓	
Observations	1,149	

Sample restricted to 10 bipartisan ministries. Poisson drops outlet–ministry cells with all-zero outcomes; $\exp(1.169) - 1 \approx 222\%$. Lee bounds identify the average treatment effect on the conditional intensive margin under a monotonicity assumption (Lee, 2009). SE clustered at the newspaper level. ** $p < 0.05$, * $p < 0.1$.

C.3 Event Studies

Figure C1: Event Study: Differential Government Advertising by Outcome



This figure presents event study estimates for all six specifications reported in Table 3. All panels include outlet–ministry or outlet–campaign fixed effects. Coefficients represent the interaction between event time (years relative to the December 2018 government transition) and outlet ideology. Error bars show 95% confidence intervals with standard errors clustered at the outlet level.

D Robustness: Low-Cost Titles

The outlets the cost-efficiency benchmark most wants to buy are those with the lowest registry rate-card cost per thousand readers in the campaign's stated demographic, and the three cheapest by that measure are high-circulation popular and sports titles, namely La Prensa at \$13.5, Esto at \$17.4 and Ovaciones at \$19.0, against \$77.3 for El Sol de México and \$113 for El Universal. Two concerns follow. Rate cards for these titles may understate the prices at which the government could actually transact, and registry audience shares may overstate how far a campaign placed in a sports paper reaches the demographic it names. Either would inflate the measured gap between the observed allocation and the benchmark.

Table D1 re-estimates equation 4 dropping these outlets. Column (2) removes the two sports titles and column (3) removes all three cheapest. Dropping an outlet removes it from the choice set, so both benchmarks are re-solved on the retained outlets and each campaign's budget is reallocated only among them. Every estimate rises rather than falls. On the cost-efficiency benchmark the binary estimate goes from \$21,609 in the full sample to \$24,261 and \$25,924, and the continuous estimate from \$28,897 to \$29,458 and \$30,199; the voter-targeting estimates move in the same direction, from \$23,932 to \$27,385 and \$30,265 on the binary measure. Standard errors on the binary specifications widen as outlets are dropped, which is expected with eighteen clusters, but the continuous specifications remain significant at the one percent level throughout. The result is not produced by the benchmark's preference for cheap high-circulation titles.

Table D1: Deviations from the Benchmarks Excluding Low-Cost Titles

	Full sample (1)	Excl. sports titles (2)	Excl. three cheapest (3)
Cost-efficiency, binary alignment	21,609** (8,991)	24,261** (10,665)	25,924** (12,022)
Cost-efficiency, continuous score	28,897*** (5,651)	29,458*** (5,553)	30,199*** (5,371)
Voter-targeting, binary alignment	23,932** (9,808)	27,385** (12,070)	30,265** (13,799)
Voter-targeting, continuous score	31,737*** (6,547)	32,875*** (6,537)	34,388*** (6,278)
Ad Campaign FE	✓	✓	✓
Outlet FE	✓	✓	✓
Observations	6,711	5,945	5,562

*Intensive-margin estimates of equation 4 in 2022 USD, at $\phi = 0.10$. Column (2) excludes Esto and Ovaciones; column (3) additionally excludes La Prensa. Both benchmarks are re-solved campaign by campaign on the retained outlets. SE clustered by outlet. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

E Prices, Placement, and the Source of the Reach Gap

Ministries pay more per unit of space than private advertisers do, and this appendix asks what that gap is and whether it drives the reach shortfall in Section 5. Measured against the price private buyers pay for the same space, a ministry pays 1.56 times as much at the median and 1.66 times as much in aggregate. Two

explanations compete. Ministries may buy better placements, since a front page in color costs more than an inside page in black and white and the market price averages across both. Or they may pay more for the same product.

Placement cannot account for a gap of that size. Outlets quote their own rate cards by position and by color, and Table E1 reports what each combination costs relative to the median quote. The most expensive product available, an odd page in color, costs 1.13 times the median, and the cheapest, an even page in black and white, costs 0.57 times. A ministry that bought nothing but the single most expensive placement on every card would therefore pay 13 percent above the middle of the market, against the 56 percent it actually pays. Premium buying explains at most a quarter of the gap.

Table E1: **Rate-Card Price per Square Centimeter by Position and Color**

	Quotes	Price per cm ² (median)	Relative to the median quote
No fixed position, colour	213	116.2	1.13×
Odd page, colour	90	114.3	1.12×
Unspecified, colour	1167	108.9	1.06×
Even page, colour	180	94.2	0.92×
No fixed position, black and white	112	84.6	0.83×
Unspecified, black and white	551	75.9	0.74×
Odd page, black and white	40	64.4	0.63×
Even page, black and white	92	58.2	0.57×

Quotes drawn from the outlets' own published tariffs, collected from commercial sites and archived copies, restricted to those reporting a positive price per square centimeter. The final column divides each combination's median by the median across all quotes. Position labels follow the cards: odd and even refer to the page on which an advertisement runs, and no fixed position means the outlet places it at its discretion.

The remaining gap is concentrated where the recorded number of advertisements is least trustworthy. A cell recording one advertisement carries a median price ratio of 4.08 and a ninetieth percentile of 27.2, while a cell recording more than twenty carries a median of 0.96 and a ninetieth percentile of 1.72. Ministries buying at scale therefore pay almost exactly the private market rate. Single-advertisement cells hold 6.6 percent of the advertisements but 34 percent of the spending, which is what a systematic undercount of insertions would produce and what a genuine price premium would not: no plausible placement costs twenty-seven times its market rate.

That distinction determines how much of the reach gap overpayment can explain, and Table E2 shows it explains almost none. Across all cells, pricing a ministry's own outlet choices at market rates would recover 96,588 readers per campaign. Restricting to cells recording at least three advertisements, where the price ratio approaches one, that figure falls to 4,422, and restricting to six or more it turns slightly negative. The misplacement term moves in the opposite direction and stays close to half a million readers throughout.

Table E2: **Reach Gap Decomposition by Number of Advertisements Recorded**

	Delivered (readers)	Overpayment (readers)	Misplacement (readers)
All cells	178,486	96,588	489,441
Cells with 3 or more	136,565	4,422	514,021
Cells with 6 or more	101,530	-6,997	520,107

Readers per campaign at $\phi = 0.10$, averaged across campaigns. Delivered reach uses the recorded number of advertisements. The overpayment column prices a ministry's own outlet choices at the private market rate and reports the additional readers that would follow. The misplacement column moves the same money to the outlets that reach the named audience most cheaply. Rows restrict the sample to cells recording at least the stated number of advertisements, retaining all cells with none.

The misplacement term is the one that does not depend on the insertion count. Both quantities it compares, what a ministry's own choices deliver at market prices and what the benchmark delivers, convert spending into insertions by dividing by the same price, so an undercount cancels. Ministries reach 275,073 readers with the outlets they chose and would reach 764,514 with the outlets that serve the campaign's audience, a factor of 2.8. That comparison stands whatever the recorded advertisement counts turn out to be.

F Can Reader Demographics Rationalize the Allocation?

The benchmark treats every reader in a campaign's named groups as equally valuable. Gentzkow et al. (2024) instead allow each viewer an intrinsic value to advertisers, which raises the possibility that the government weights demographic groups differently and that the apparent misallocation reflects those weights rather than ideology. This appendix asks what weights would be required.

Replace the audience with $a_i(w) = N_i \sum_g w_g r_{ig}$, where r_{ig} is outlet i 's share of readers in group g across three income and six age categories, and choose $w \geq 0$ to minimize the squared distance between the resulting optimal allocation and the observed one.

The weights are not identified, for a reason that is exact rather than numerical. Income shares sum to one and age shares sum to one for every outlet, so adding a constant to all three income weights and subtracting the same constant from all six age weights leaves $a_i(w)$ unchanged for every outlet, verified to machine precision at any magnitude. The objective is correspondingly flat: fixing any single group's weight at zero and re-optimizing the rest costs at most 2.7 percent of the objective, and seven of the nine cost under half a percent. A further obstacle is that r_{ig} is time-invariant and outlet-specific, so $a_i(w)$ does not vary across campaigns and cannot in principle reproduce campaign-specific targeting. The estimated weights are therefore not reported.

What the exercise does deliver is a comparison. Table F1 reports how well each specification reproduces the observed allocation. The informative column is the last, the correlation between predicted and observed *within-campaign* shares, since the correlation in levels is largely driven by campaign budget. Splitting each budget equally across outlets, with no model at all, attains a level correlation of 0.49. Every audience-based

specification predicts within-campaign allocation negatively. A single parameter loading circulation on outlet ideology reverses that, and outperforms nine free demographic weights.

Table F1: Fit of Alternative Audience Weightings to the Observed Allocation

	Objective ($\times 10^{13}$) (1)	Correlation, levels (2)	Correlation, within-campaign shares (3)
Equal weights, $a_i \propto N_i$	9.604	0.453	-0.103
Benchmark audience weight	9.627	0.454	-0.083
Best demographic weights (9 parameters)	9.273	0.479	-0.056
Circulation $\times$ ideology (1 parameter)	9.065	0.505	+0.110
Demographic weights + ideology	8.954	0.508	+0.062

Each row solves equation 2 campaign by campaign with the stated audience weight and compares the result to observed spending. The objective is the summed squared deviation across outlet-campaign cells. The ideology specification sets $a_i = N_i \exp(\theta \cdot \text{score}_i)$ with θ estimated freely; the fitted value is $\theta = 1.28$. Column (3) computes the correlation of predicted and observed spending shares within each campaign, then averages.

G Robustness: The Exposure Rate ϕ

The technology of Section 5 has one free parameter, the probability ϕ that a reader sees any one insertion. Rather than defend a value, I report the estimate across the range over which the budget constraint binds. As $\phi \rightarrow 0$ reach becomes linear in spending and the optimum is a corner placing the entire budget in the cheapest outlet per target reader, the maximum-concentration allocation. The upper end is degenerate and is therefore not a comparable column. As $\phi \rightarrow 1$ a single insertion saturates an outlet, so additional spending buys no further reach and the budget ceases to bind. The corresponding allocation spends only 57 percent of the average campaign budget and underspends in 251 of 383 campaigns, so a dollar deviation measured against it would mechanically contain the unspent remainder rather than a reallocation.

Table G1: Intensive-Margin Deviation Across the Full Range of ϕ

	$\phi \rightarrow 0$ (corner) (1)	$\phi = 0.05$ (2)	$\phi = 0.10$ (3)	$\phi = 0.15$ (4)
Aligned (binary)	38,608 (35,590)	24,303*** (8,051)	21,609** (8,991)	20,168** (9,427)
Aligned (continuous)	43,704** (17,491)	29,090*** (4,867)	28,897*** (5,651)	28,621*** (5,981)
Ad Campaign FE	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711

Deviations from the cost-efficiency benchmark, in 2022 USD. SE clustered by outlet.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column (1) is the $\phi \rightarrow 0$ corner and columns (2) to (4) are interior. Column (3) reproduces Panel B, column (1) of Table 4.

Table G2 reports the distribution across campaigns of the budget multiplier λ recovered in Section 5, of the reservation cost per target reader it implies, and of the rate-card costs those are compared against. The reservation cost sits inside the observed price distribution at every quantile, which is what produces interior solutions rather than corner ones.

Table G2: Recovered Shadow Price and Reservation Cost Across Campaigns

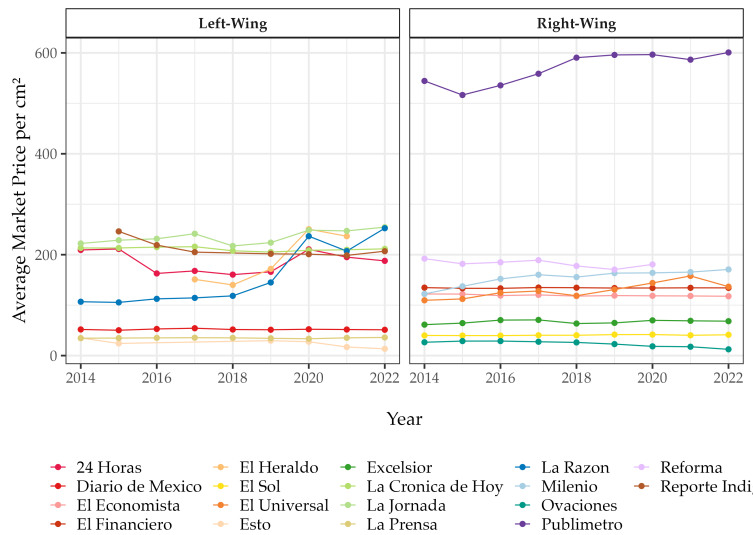
	p10	p25	Median	p75	p90
Shadow price λ (target readers per USD)	0.15	0.30	0.59	1.12	3.12
Reservation cost $\bar{c}$ (USD per 1,000 target readers)	34	94	177	353	727
Rate-card cost per 1,000 target readers	18	96	161	251	350

One observation per campaign, at $\phi = 0.10$. λ is recovered by bisection on the budget constraint under equation 3. The reservation cost is $\bar{c} = \kappa/\lambda$ with $\kappa = -\ln(1 - \phi)$. The last row is the registry rate-card cost per thousand readers in the campaign's stated demographic, pooled across outlet-campaign cells.

The intensive-margin deviation is positive at every point in the bracket, from \$38,608 at the concentration limit to \$20,168 at $\phi = 0.15$, and the continuous-score specification is significant at the five percent level at the corner and at the one percent level throughout the interior. The calibration of ϕ therefore does no work in the conclusion. Two quantities are more sensitive and should be read with that in mind. The number of outlets the efficient allocation uses rises from one at the concentration limit to 10.3 at $\phi = 0.10$ and 15.4 at the coverage limit, so the statement that campaigns are more concentrated than efficiency requires is qualitative rather than a point estimate. And the foregone-audience gap ranges from roughly three to five times delivered reach across the interior of the bracket.

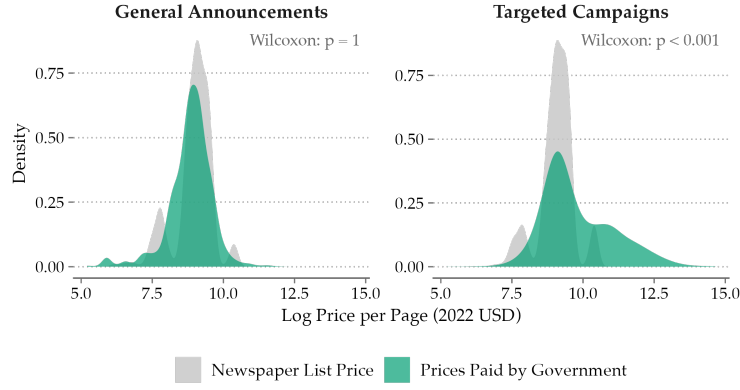
H Advertising Market Prices

Figure H1: Market Prices for Print Advertising by Outlet (2014–2022)



This figure displays the average market price per cm^2 for print advertisements across 18 newspaper outlets in Mexico from 2014 to 2022. Market price is constructed by first calculating the price per cm^2 for each individual advertisement, then averaging across all advertisements within each outlet-year.

Figure H2: Distribution of Government and Market Prices by Type of Campaign (2014–2022)



This figure compares the distribution of prices paid by the government for advertising contracts with the corresponding market prices by the type of communication campaign. Government prices are observed transaction prices from government advertising contracts. Market prices represent the average price per cm² for the corresponding outlet, multiplied by the advertisement size in the government contract.

I Reader Sorting and the Voter-Targeting Benchmark

The voter-targeting benchmark in Section 5 builds copartisan reach from where an outlet circulates, so it misses any tendency of incumbent supporters to choose newspapers that share their politics. This appendix asks how strong that sorting would have to be before the result disappears. Writing ρ_{it} for the measured copartisan share, I tilt it by

$$\text{logit}(\tilde{\rho}_{it}) = \text{logit}(\rho_{it}) + \tau z_i, \quad z_i = \frac{1}{2} \text{ if aligned, } -\frac{1}{2} \text{ otherwise} \quad (6)$$

so that e^τ scales the odds that an incumbent supporter reads an aligned rather than a misaligned newspaper. I apply the tilt in log-odds so that it means the same thing at any base rate, which matters because the incumbent's vote share differs sharply across the two administrations. I then rebuild the voter-targeting benchmark with $\tilde{\rho}_{it}$ and estimate equation 4 again at each value.

Table I1: Deviations from the Voter-Targeting Benchmark Under Assumed Reader Sorting

	Assumed sorting, odds ratio e^{τ}					
	1.00 (1)	1.28 (2)	1.65 (3)	2.12 (4)	2.72 (5)	4.48 (6)
<i>Panel A. Deviation in 2022 dollars</i>						
Aligned	23,932** (9,808)	17,365* (9,365)	10,806 (8,999)	4,549 (8,720)	-1,329 (8,532)	-11,585 (8,364)
Ideology score	31,737*** (6,547)	27,095*** (6,660)	22,464*** (6,948)	18,063** (7,306)	13,988* (7,712)	6,991 (8,536)
<i>Panel B. Deviation as share of campaign budget (pp)</i>						
Aligned	3.40** (1.49)	1.98 (1.37)	0.54 (1.30)	-0.78 (1.27)	-1.98 (1.28)	-4.16*** (1.36)
Ideology score	4.11** (1.41)	3.14* (1.50)	2.17 (1.66)	1.29 (1.82)	0.51 (1.99)	-0.82 (2.36)
Implied aligned-misaligned gap (pp)	-0.9	+3.2	+7.4	+11.5	+15.6	+23.8
Ad Campaign FE	✓	✓	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711	6,711	6,711

Intensive-margin estimates of equation 4, with the copartisan share replaced by $\bar{p}_{it}$ from equation 6. Panel A takes the deviation in 2022 dollars and Panel B divides it by the campaign's total budget. Column (1) uses the measured share. The measured baseline odds ratio does not equal one, so the headings report the assumed multiplier applied to it rather than the resulting ratio. The final row reports the implied difference in mean copartisan share between aligned and misaligned outlets, which depends jointly on the multiplier and on the base rate. Standard errors clustered by outlet. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

How much sorting the result tolerates depends on the scale I measure the deviation on. In Table I1 I raise the assumed sorting from none in column (1) to a multiplier of 4.5 in column (6), rebuilding the voter-targeting benchmark and re-estimating equation 4 at each step. Panel A measures the deviation in dollars and survives the most sorting, since the continuous estimate stays significant at the five percent level out to a multiplier of 2.1 and at ten percent out to 2.7. Panel B divides each deviation by its campaign budget and survives less, since there the continuous estimate holds at ten percent only to 1.3 and the binary estimate loses significance as soon as I impose any sorting at all. The binary estimate turns negative in both panels once the multiplier passes 2.1, and by 4.5 the sign reverses at the one percent level in Panel B, so sorting that strong would make the benchmark want to buy aligned newspapers more heavily than ministries did. Taking the share scale as the binding one, the result survives as long as the odds of an incumbent supporter reading an aligned rather than a misaligned newspaper stay within roughly 1.3 times what I measure, which would put 24 percent of an aligned outlet's readers on the incumbent's side against 21 percent of a misaligned outlet's.

J Estimating Outlet-Level Copartisan Shares

I measure each outlet's copartisan reach using geographic variation in circulation. For each outlet j , year t , and state s , I compute the weight w_{js} as outlet j 's share of its own total circulation that falls in state s :

$$w_{js} = \frac{\text{circ}_{jst}}{\sum_{s'} \text{circ}_{js't}} \quad (7)$$

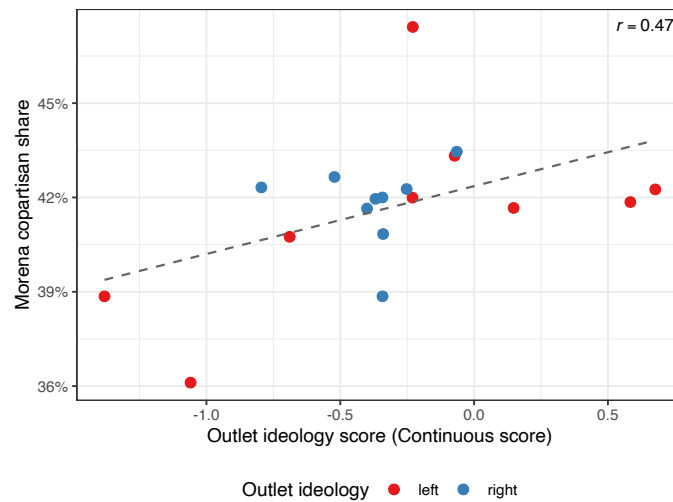
The copartisan share for party p in election year τ is then the circulation-weighted average of state-level vote shares:

$$\widehat{\text{Copartisan}}_{j\tau}^p = \sum_s w_{jst} \cdot v_{s\tau}^p \quad (8)$$

where $v_{s\tau}^p$ is the official vote share of party p in state s at election τ . Circulation data come from a balanced panel of regional edition figures for 18 national newspapers.¹⁰

A key assumption underlying the voter-targeting benchmark is that Morena and PRI governments could, in principle, select outlets whose readership skews toward their own supporters. Figure J1 tests this by plotting each outlet's ideology score, a continuous measure of the partisan valence of its editorial language (Gentzkow et al., 2019), against its estimated Morena copartisan share in 2018. The ideological gap in copartisan reach is negligible. Left-wing and right-wing outlets attract nearly identical partisan audiences, and all 18 outlets fall within a 10-percentage-point band. The average Morena copartisan share is 41.6% among left-wing outlets and 41.8% among right-wing outlets. Voters in Mexico do not sort into newspapers by ideology, which limits the scope for outlet-level partisan targeting.

Figure J1: Outlet Ideology Score and Morena Copartisan Share (2018)



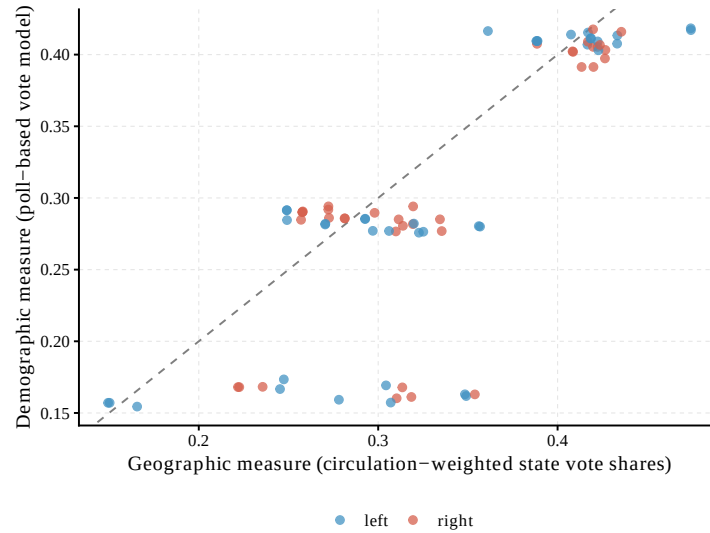
Each point is one of 18 national newspapers. The x -axis is the outlet's continuous ideology score from Gentzkow et al. (2019); more negative values indicate more left-leaning editorial language. The y -axis is the estimated share of the outlet's readers who supported Morena, computed as a circulation-weighted average of state-level Morena vote shares (2018 presidential election). The dashed line is an OLS fit. r denotes the Pearson correlation.

The geographic measure can be checked against an independent construction. Individual polling microdata registered with the National Electoral Institute records vote intention alongside state, gender, education, age and income; I estimate vote choice on those covariates and apply the fitted model to each outlet's registry audience profile, giving a demographic estimate of copartisan reach that shares no inputs with the geographic measure beyond state of residence. Figure J2 plots the two against each other across the 76 outlet-years where both are available. They correlate at 0.80. On the demographic measure left-wing

¹⁰Circulation is measured as copies per 10,000 inhabitants, obtained from the Padrón Nacional de Medios Impresos (PNMI). State-level vote shares come from official results published by the Instituto Nacional Electoral (INE) for the 2015, 2018, and 2021 midterm and presidential elections. Because advertising campaigns span multiple calendar years, I assign each year to the nearest election cycle: 2014–2016 to the 2015 midterm, 2017–2019 to the 2018 presidential election, and 2020–2022 to the 2021 midterm.

outlets reach a Morena share of 30.1 percent against 30.5 percent for right-wing outlets, a gap of -0.4 percentage points; on the geographic measure the figures are 33.3 and 33.4 percent, a gap of -0.2 points. The absence of an ideological gap in copartisan reach is therefore not an artifact of building the measure from circulation geography.

Figure J2: Demographic and Geographic Estimates of Copartisan Reach

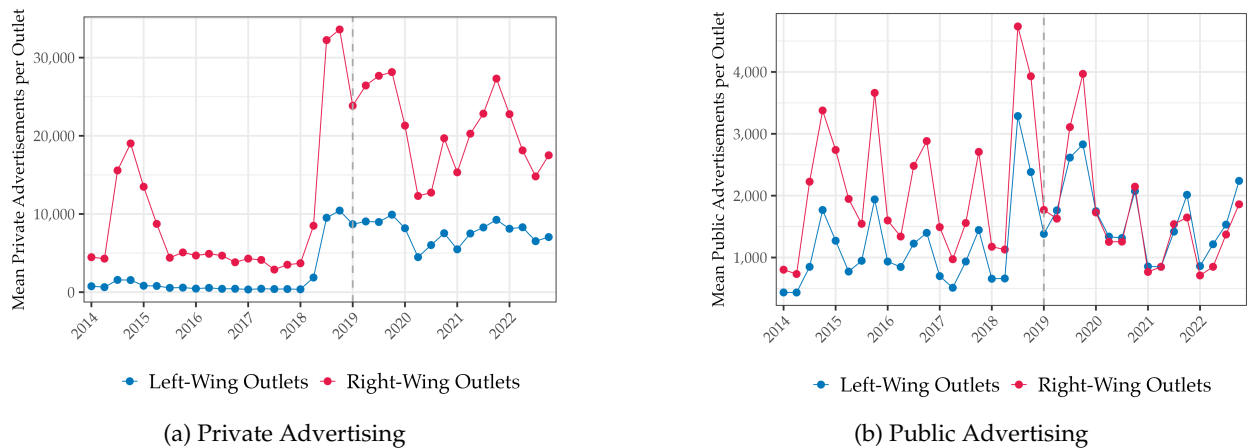


Estimated Morena share of each outlet's readership, 76 outlet-years. The horizontal axis weights state-level official vote shares by the outlet's circulation across states. The vertical axis applies a vote-choice model estimated on polling microdata to the outlet's registry audience profile. The dashed line is the 45-degree line; colors denote the outlet's ideology classification from Section 3.2.

K Downstream Effects in the Media

K.1 Revenue Shock to Outlets

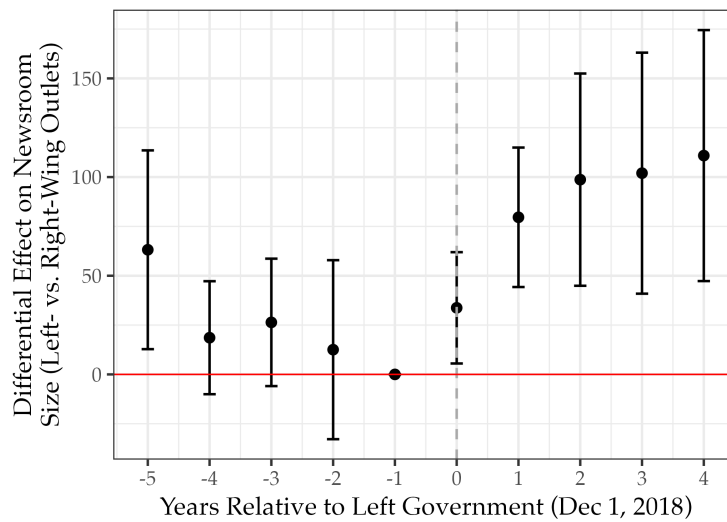
Figure K1: Quarterly Advertising Trends by Outlet Ideology (2014–2022)



The figure displays the mean number of advertisements per outlet by quarter, averaged separately across 9 left-wing and 9 right-wing newspaper outlets. Advertisements are classified as private sector or public sector based on a dictionary-based method. The vertical dashed line marks the first quarter of 2019.

K.2 Newsroom Size

Figure K2: Event Study: Differential Effect of Left-Wing Government on Newsroom Size



This figure presents event study estimates of the differential effect of the left-wing government transition on newsroom size. The specification corresponds to Column (1) in Table 5. Event time -1 is normalized to zero as the reference period. Error bars indicate 95% confidence intervals. The specification includes outlet and year fixed effects, with standard errors clustered at the outlet level.

Table K1: Effect of Left-Wing Government on Newsroom Size (Levels)

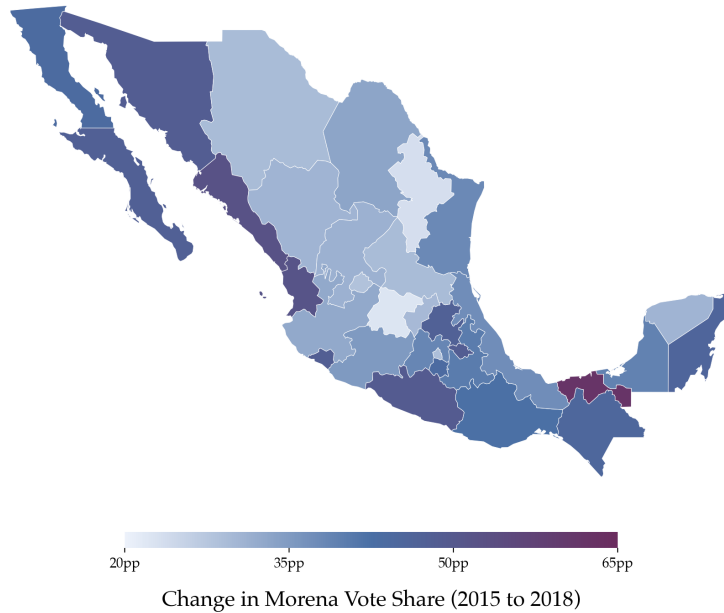
	<i>Dependent variable:</i>		
	Total Newsroom	Staff	Journalists
	(1)	(2)	(3)
Left Gov × Left Outlet	67.683*** (13.804)	38.868*** (12.842)	28.567* (16.287)
Mean pre-2018 (Left)	139.73	102	37.95
Mean pre-2018 (Right)	251.54	186.96	64.58
SD pre-2018	160.96	124.93	41.49
Year FE	✓	✓	✓
Outlet FE	✓	✓	✓
Observations	1,908	1,908	1,908
R ²	0.946	0.960	0.742

Table K2: News Production by Byline and Article Type

	IHS(Total	IHS(Avg	Articles by Byline Type (IHS)			Articles by Type (IHS)	
	Articles)	Length)	Newsroom	Agency	House	Hard	Soft
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Left- Gov $\times$ Left-Wing Outlet	0.405** (0.205)	0.128 (0.079)	0.351 (0.250)	0.477 (0.391)	0.412** (0.184)	0.321 (0.256)	0.451** (0.208)
Mean Outcome (Pre-2018, Left Outlet)	1686.64	191.05	874.63	267.22	542.56	690.69	183.94
Mean Outcome (Pre-2018, Right Outlet)	3079.95	201.45	1736.33	421.64	895.18	1374.98	361.34
SD Outcome (Pre-2018)	1511.03	39.21	902.49	280.79	505.89	747.07	196.76
Year FE	✓	✓	✓	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓	✓	✓	✓
Observations	1,908	1,908	1,908	1,908	1,908	1,908	1,908
R ²	0.559	0.200	0.709	0.642	0.490	0.766	0.750

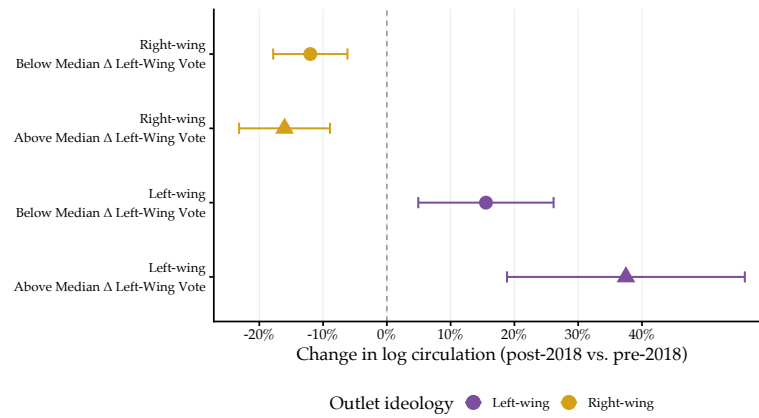
K.3 Circulation

Figure K3: Change in Morena Vote Share by State (2015–2018)



This figure maps the change in Morena's vote share by state between the 2015 midterm and the 2018 presidential election. Darker shades indicate larger shifts toward Morena. The dashed line in the triple difference-in-differences specification classifies states above the sample median shift ($D_s = 1$) as high-shift states. The variation ranges from approximately 20 percentage points in the north and northwest to over 65 percentage points in the south and southeast.

Figure K4: Content-Driven vs. Preference-Driven: Coefficient Plot



This figure displays estimated circulation changes by outlet ideology and state-level political shift. Left-wing outlets in low-shift states capture the content-driven component; the additional gain for left-wing outlets in high-shift states captures the preference-driven component.